Training Neural Networks

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BY NC SA
unless otherwise stated

Credits

Deep Learning courses (NPFL114, NPFL138) by Milan Straka

- https://ufal.mff.cuni.cz/courses/npfl114, https://ufal.mff.cuni.cz/courses/npfl138

The Deep Learning book by Ian Goodfellow, Yoshua Bengio, and Aaron Courville

- http://www.deeplearningbook.org

The Dive into Deep Learning book by Aston Zhang, Zachary C. Lipton, Mu Li, and Alexander J. Smola

- http://www.d2l.ai

Arxiv.org on-line repository for most-recent papers from AI

- https://arxiv.org

Refresh – Neural Networks

- Neural network describes a computation, which gets an input tensor and produces an output.
  - For the time being, the input tensor has a fixed size.
  - The input tensor is usually a vector, but it can be 2D/3D/4D tensor.
    - images, video, time sequences like speech, ...
  - The output usually describes a distribution.
    - normal distribution for regression
    - Bernoulli for binary classification
    - categorical for multiclass classification
- The basic units are nodes, composed in an acyclic graph.
- The edges have weights, nodes have activation functions.
- Nodes of neural networks are usually composed in layers.

Machine Learning Basics

Machine Learning Basics

We usually have a **training set**, which is assumed to consist of examples generated independently from a **data-generating distribution**.

The goal of *optimization* is to match the training set as well as possible.

However, the goal of *machine learning* is to perform well on *previously unseen* data, to achieve lowest **generalization error** or **test error**. We typically estimate it using a **test set** of examples **independent of the training set**, but **generated by the same data generating distribution**.

According to the **No free lunch theorem** (Wolpert, 1995 & 1997), every classification algorithm achieves the same *overall* error when processing unseen examples from *all possible* (independent) data distributions for **training set** and **test set**.

According to the **No free lunch theorem** for computer complexity and optimization, no machine learning algorithm is *universally* better than others if the optimized function f has no other underlying structure (e.g., a differentiable function).

But in practice the data distributions are not uniformly random and the training set and the test set are somehow dependent, thus some algorithms might work better than others.

Machine Learning Basics

Challenges in machine learning:

- *underfitting* (the model is "too weak", bad performance even on training set)
- *overfitting* (the model is "too strong", learned rules are too specific and do not generalize)

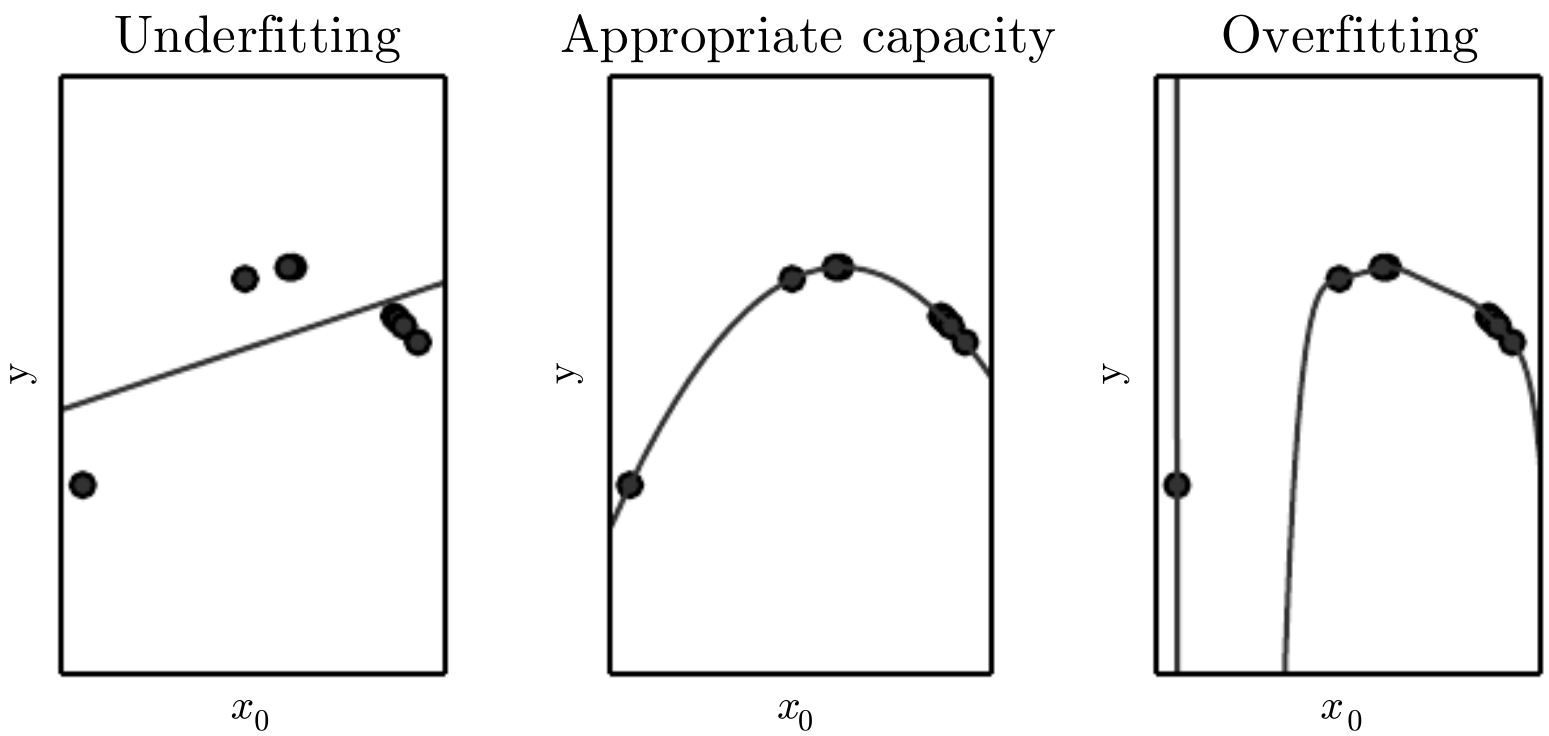
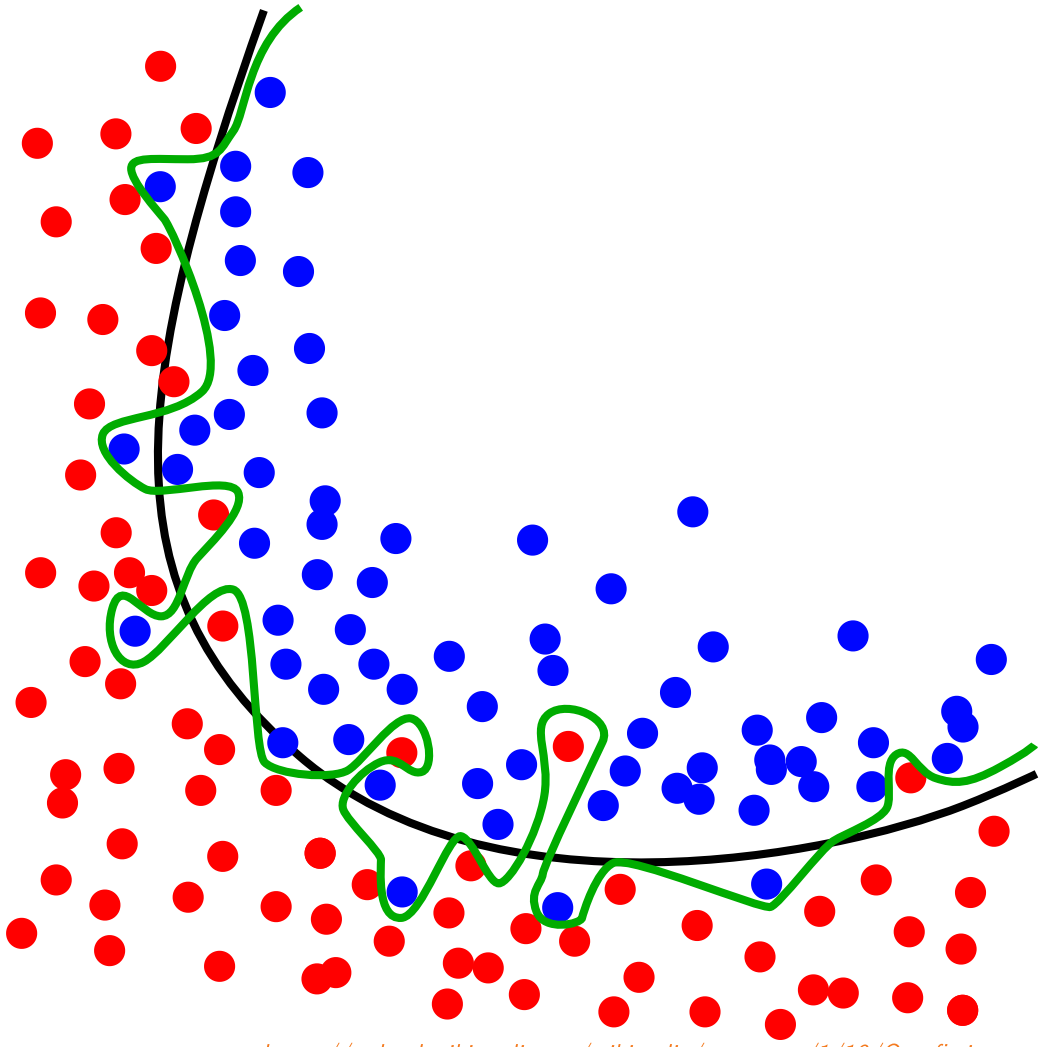


Figure 5.2 of "Deep Learning" book, https://www.deeplearningbook.org

Machine Learning Basics



https://upload.wikimedia.org/wikipedia/commons/1/19/Overfitting.svg

Machine Learning Basics

We can control whether a model underfits or overfits by modifying its *capacity*.

- *representational capacity* (what the model could represent, depends on the model size)
- *effective capacity* (what the model actually learns, depends on training, regularization, ...)

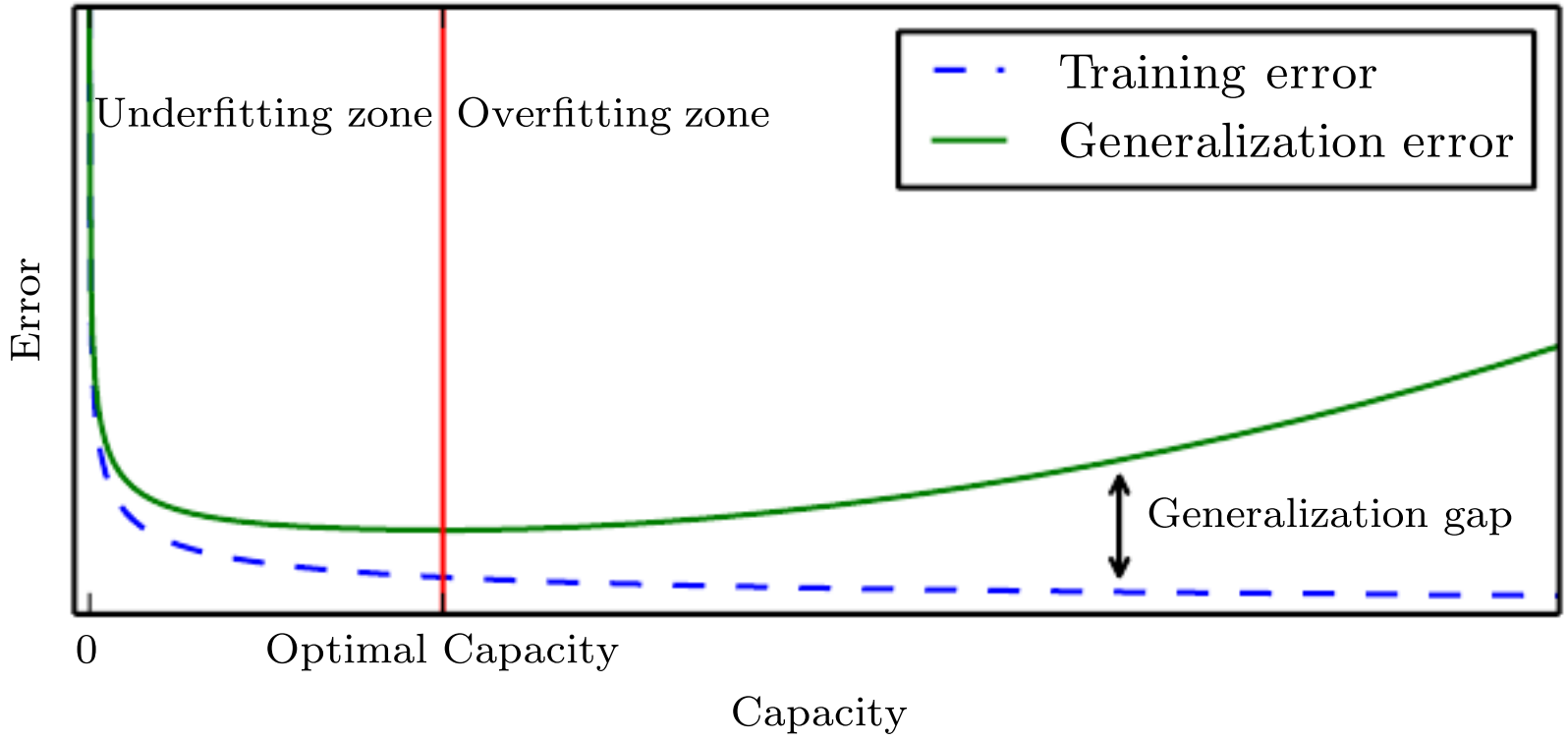


Figure 5.3 of "Deep Learning" book, https://www.deeplearningbook.org

Machine Learning Basics

Overfitting usually decreases with the amount of the training data.

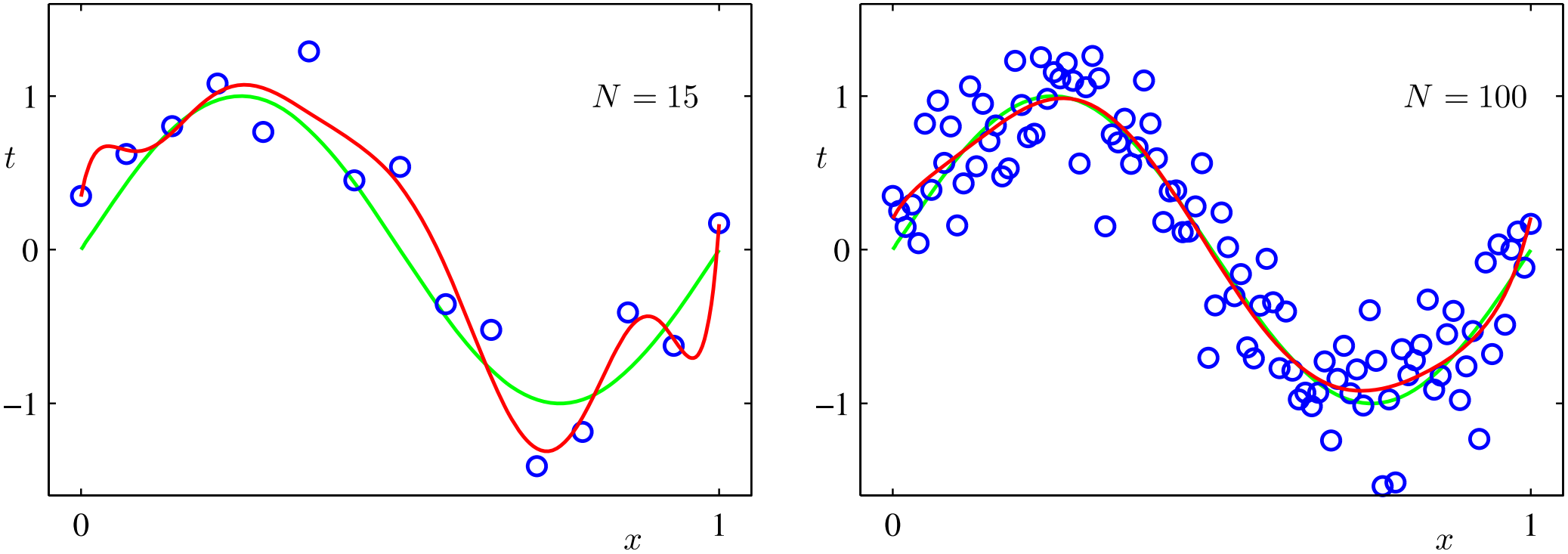


Figure 1.6 of Pattern Recognition and Machine Learning.

Machine Learning Basics

Any change in a machine learning algorithm that is designed to *reduce generalization error* (but not necessarily its training error) is called **regularization**.

L^2 **regularization** (also called **weight decay**) penalizes models with large weights (using a penalty of $\frac{1}{2}\|\boldsymbol{\theta}\|^2$).

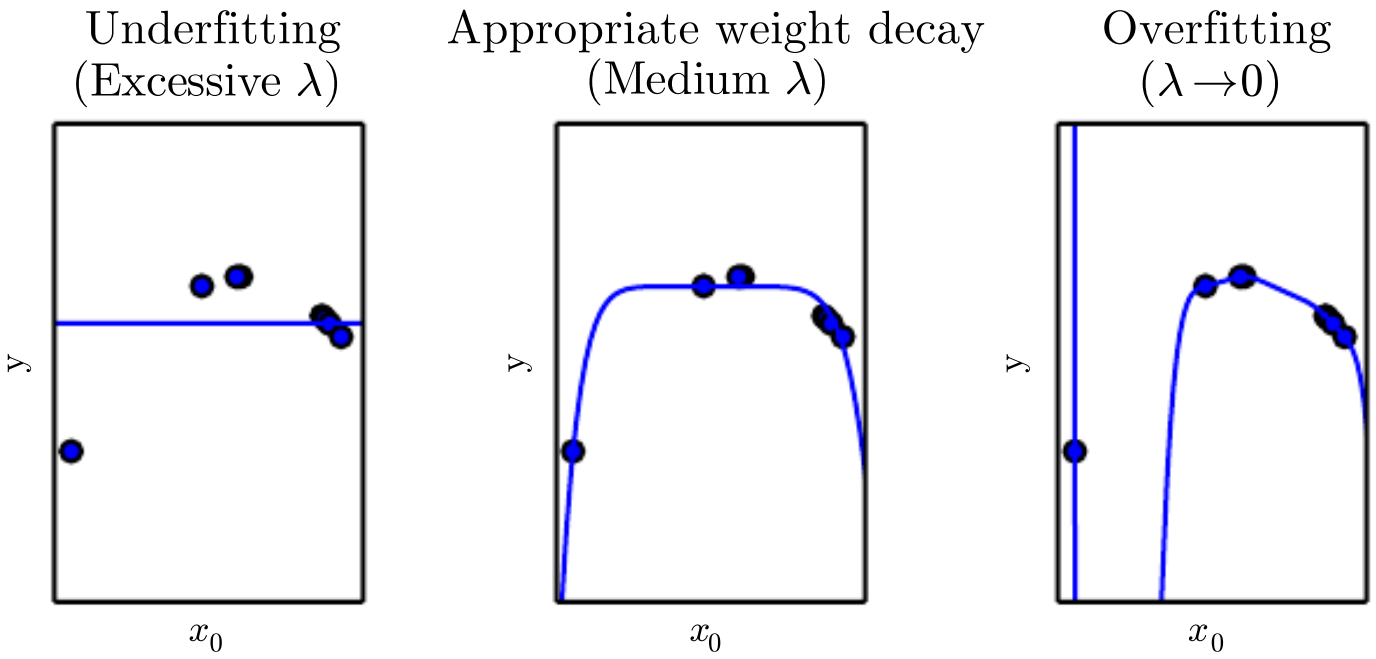


Figure 5.5 of "Deep Learning" book, https://www.deeplearningbook.org

Machine Learning Basics

Hyperparameters are not adapted by a learning algorithm itself, while the model **parameters** (weights, biases) are adapted by it.

Usually a **development set**, also called a **validation set**, is used to estimate the generalization error, allowing to update hyperparameters accordingly.

Maximum Likelihood Estimation

Self-Information

Self-information can be considered the amount of **surprise** when a random variable is sampled.

- Should be zero for events with probability 1.
- Less likely events are more surprising.
- Independent events should have **additive** surprise (information).

These conditions are fulfilled by self-information $I(x)$, also called surprise:

$$I(x) \stackrel{\text{def}}{=} -\log P(x) = \log \frac{1}{P(x)}.$$

Entropy

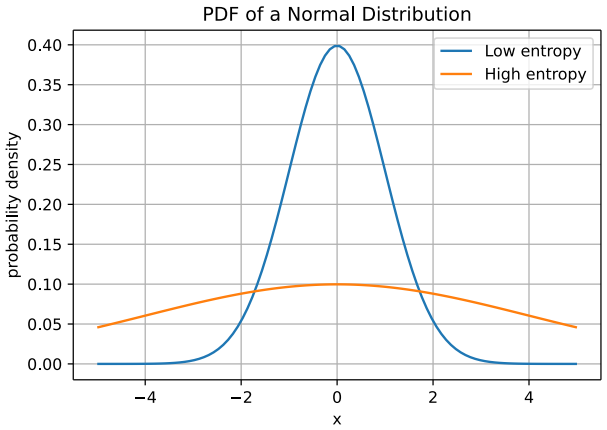
Amount of **surprise** in the whole distribution.

$$H(P) \stackrel{\text{def}}{=} \mathbb{E}_{\mathbf{x}\sim P}[I(x)] = -\mathbb{E}_{\mathbf{x}\sim P}[\log P(x)]$$

In order to encode data drawn randomly from the distribution P, we need at least $H(P)$ of bits ("nats") on average to encode it (Shannon 1948)

- for discrete P: $H(P) = -\sum_x P(x) \log P(x)$
- for continuous P: $H(P) = -\int P(x) \log P(x)\, \mathrm{d}x$

(For $P(x) = 0$ we consider $P(x) \log P(x)$ to be zero.)



Note that in the continuous case, the continuous entropy (also called *differential entropy*) has slightly different semantics, for example, it can be negative.

From now on, all logarithms are *natural logarithms* with base e.

Entropy

Amount of **surprise** in the whole distribution.

$$H(P) \stackrel{\text{def}}{=} \mathbb{E}_{\mathbf{x}\sim P}[I(x)] = -\mathbb{E}_{\mathbf{x}\sim P}[\log P(x)]$$

Entropy *shows the average number of bits (nats) required to transmit a randomly selected event from a probability distribution.*

https://machinelearningmastery.com/cross-entropy-for-machine-learning/

Cross-Entropy

$$H(P,Q) \stackrel{\text{def}}{=} -\mathbb{E}_{\mathbf{x}\sim P}[\log Q(x)]$$

- generally $H(P,Q) \neq H(Q,P)$
- Gibbs inequality
  - $H(P,Q) \geq H(P)$
  - $H(P) = H(P,Q) \Leftrightarrow P = Q$
  - Corollary: For a categorical distribution with n outcomes, $H(P) \leq \log n$.
- conditional (cross-)entropy (useful for maximum likelihood estimation) is defined as $H(Y|X) = H(X,Y) - H(X)$

*The **cross entropy** is the average number of bits (nats) needed to encode data coming from a real data source distribution P using model Q.*

https://machinelearningmastery.com/cross-entropy-for-machine-learning/

Kullback-Leibler Divergence (KL Divergence)

Sometimes also called **relative entropy**.

$$D_{\mathrm{KL}}(P\|Q) \stackrel{\text{def}}{=} H(P,Q) - H(P)$$

- consequence of Gibbs inequality: $D_{\mathrm{KL}}(P\|Q) \geq 0$, $D_{\mathrm{KL}}(P\|Q) = 0$ iff $P = Q$
- generally $D_{\mathrm{KL}}(P\|Q) \neq D_{\mathrm{KL}}(Q\|P)$

*The **KL divergence** is the average number of **extra** bits (nats) needed to encode the data, due to the fact that we used distribution Q to encode the data (model) instead of the real data source distribution P.*

https://machinelearningmastery.com/cross-entropy-for-machine-learning/

Cross-Entropy (and KL Divergence) Example

Training sample 1

y_true distribution P	Y_pred distribution Q	$-p_i \cdot \ln(q_i)$
0	0.05	0
1	0.95	0.051
0	*0**	*0**
	Σ	0.051

Training sample 2

y_true distribution P	Y_pred distribution Q	$-p_i \cdot \ln(q_i)$
0	0.1	0
0	0.8	0
1	0.1	2.303
	Σ	2.303

*zero values in Q should be somehow fixed or avoided

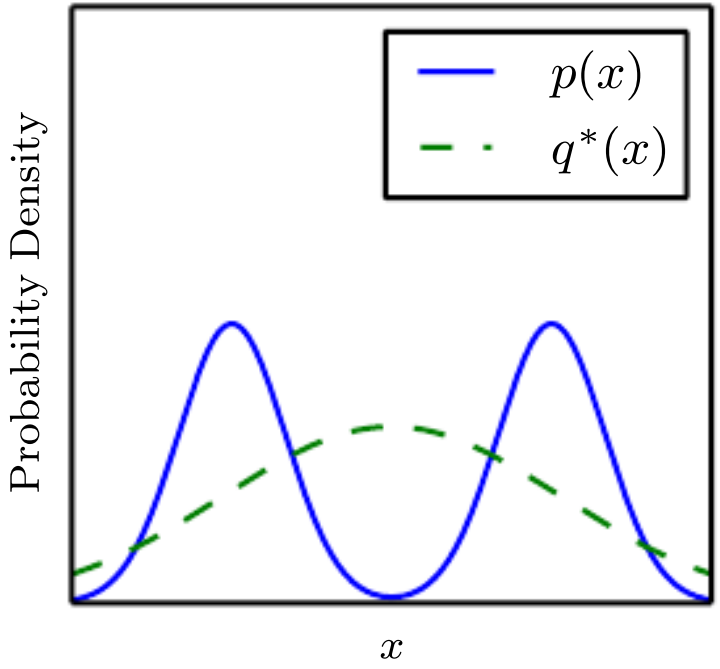
Average cross-entropy $H(P, Q)$ is then $(0.051 + 2.303)/2 = 1.177$.

In TensorFlow (`tf.keras.losses.CategoricalCrossentropy`), zero values in Q are treated as some very small value (e.g. 10^{-7}) and the vector is rescaled to sum to 1. In PyTorch the problem is marginalized by computing loss on logits (`torch.nn.CrossEntropyLoss`) or on logSoftmax outputs (`torch.nn.NLLLoss`), thus in PyTorch no fixes are applied.

KL Divergence

Assume we want to find the best one-peak approximation q^* of a two-peak distribution p:

$$q^* = \text{argmin}_q D_{\text{KL}}(p \| q)$$



Modified from Figure 3.6 of "Deep Learning" book, https://www.deeplearningbook.org

Loss Function

A model is usually trained in order to minimize the **loss** on the training data.

Assuming that a model computes $f(\boldsymbol{x};\boldsymbol{\theta})$ using parameters $\boldsymbol{\theta}$, the **mean square error** of given N examples $\left(\boldsymbol{x}^{(1)}, y^{(1)}\right), \left(\boldsymbol{x}^{(2)}, y^{(2)}\right), \ldots, \left(\boldsymbol{x}^{(N)}, y^{(N)}\right)$ is computed as

$$\frac{1}{N}\sum_{i=1}^{N}\left(f(\boldsymbol{x}^{(i)};\boldsymbol{\theta}) - y^{(i)}\right)^2.$$

A common principle used to design loss functions is the **maximum likelihood principle**.

Maximum Likelihood Estimation

Let $\mathbb{X} = \{\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \ldots, \boldsymbol{x}^{(N)}\}$ be training data drawn independently from the data-generating distribution p_{data}.

We denote the **empirical data distribution** as $\hat{p}_{\text{data}}$, where

$$\hat{p}_{\text{data}}(\boldsymbol{x}) \stackrel{\text{def}}{=} \frac{\left|\{i : \boldsymbol{x}^{(i)} = \boldsymbol{x}\}\right|}{N}.$$

Let $p_{\text{model}}(\mathbf{x}; \boldsymbol{\theta})$ be a family of distributions.

- If the weights are fixed, $p_{\text{model}}(\mathbf{x}; \boldsymbol{\theta})$ is a probability distribution.
- If we instead consider the fixed training data $\mathbb{X}$, then

$$L(\boldsymbol{\theta}) = p_{\text{model}}(\mathbb{X}; \boldsymbol{\theta}) = \prod_{i=1}^{N} p_{\text{model}}(\boldsymbol{x}^{(i)}; \boldsymbol{\theta})$$

  is called the **likelihood**. Note that even if the value of the likelihood is in range $[0, 1]$, it is not a probability, because the likelihood is not a probability distribution.

Maximum Likelihood Estimation

Let $\mathbb{X} = \{\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \ldots, \boldsymbol{x}^{(N)}\}$ be training data drawn independently from the data-generating distribution p_{data}. We denote the empirical data distribution as $\hat{p}_{\text{data}}$ and let $p_{\text{model}}(\mathbf{x}; \boldsymbol{\theta})$ be a family of distributions.

The **maximum likelihood estimation** of $\boldsymbol{\theta}$ is:

$$
\begin{aligned}
\boldsymbol{\theta}_{\text{MLE}} &= \arg\max_{\boldsymbol{\theta}} p_{\text{model}}(\mathbb{X}; \boldsymbol{\theta}) = \arg\max_{\boldsymbol{\theta}} \prod_{i=1}^{N} p_{\text{model}}(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}) \\
&= \arg\min_{\boldsymbol{\theta}} \sum_{i=1}^{N} -\log p_{\text{model}}(\boldsymbol{x}^{(i)}; \boldsymbol{\theta}) \\
&= \arg\min_{\boldsymbol{\theta}} \mathbb{E}_{\mathbf{x} \sim \hat{p}_{\text{data}}}[-\log p_{\text{model}}(\boldsymbol{x}; \boldsymbol{\theta})] \\
&= \arg\min_{\boldsymbol{\theta}} H(\hat{p}_{\text{data}}(\mathbf{x}), p_{\text{model}}(\mathbf{x}; \boldsymbol{\theta})) \\
&= \arg\min_{\boldsymbol{\theta}} D_{\text{KL}}(\hat{p}_{\text{data}}(\mathbf{x}) \| p_{\text{model}}(\mathbf{x}; \boldsymbol{\theta})) + H(\hat{p}_{\text{data}}(\mathbf{x}))
\end{aligned}
$$

Maximum Likelihood Estimation

MLE can be easily generalized to the conditional case, where our goal is to predict y given $\boldsymbol{x}$:

$$\begin{aligned}
\boldsymbol{\theta}_{\text{MLE}} &= \arg\max_{\boldsymbol{\theta}} p_{\text{model}}(\mathbb{Y}|\mathbb{X};\boldsymbol{\theta}) \\
&= \arg\min_{\boldsymbol{\theta}} \sum\nolimits_{i=1}^{N} -\log p_{\text{model}}(y^{(i)}|\boldsymbol{x}^{(i)};\boldsymbol{\theta}) \\
&= \arg\min_{\boldsymbol{\theta}} \mathbb{E}_{(\mathbf{x},\mathrm{y})\sim\hat{p}_{\text{data}}}[-\log p_{\text{model}}(y|\boldsymbol{x};\boldsymbol{\theta})] \\
&= \arg\min_{\boldsymbol{\theta}} H(\hat{p}_{\text{data}}(\mathrm{y}|\mathbf{x}), p_{\text{model}}(\mathrm{y}|\mathbf{x};\boldsymbol{\theta})) \\
&= \arg\min_{\boldsymbol{\theta}} D_{\text{KL}}(\hat{p}_{\text{data}}(\mathrm{y}|\mathbf{x})\|p_{\text{model}}(\mathrm{y}|\mathbf{x};\boldsymbol{\theta})) + H(\hat{p}_{\text{data}}(\mathrm{y}|\mathbf{x}))
\end{aligned}$$

where the conditional entropy is defined as $H(\hat{p}_{\text{data}}) = \mathbb{E}_{(\mathbf{x},\mathrm{y})\sim\hat{p}_{\text{data}}}[-\log(\hat{p}_{\text{data}}(y|\boldsymbol{x}))]$ and the conditional cross-entropy as $H(\hat{p}_{\text{data}}, p_{\text{model}}) = \mathbb{E}_{(\mathbf{x},\mathrm{y})\sim\hat{p}_{\text{data}}}[-\log(p_{\text{model}}(y|\boldsymbol{x};\boldsymbol{\theta}))]$.

The resulting *loss function* is called **negative log-likelihood** (**NLL**), or **cross-entropy**, or **Kullback-Leibler divergence**.

Estimators and Bias

An **estimator** is a rule for computing an estimate of a given value, often an expectation of some random value(s). For example, we might estimate *mean* of a random variable by sampling a value according to its probability distribution.

The **bias** of an estimator is the difference of the expected value of the estimator and the true value being estimated. If the bias is zero, we call the estimator **unbiased**, otherwise **biased**.

If we have a sequence of estimates, it might also happen that the bias converges to zero. Consider the well-known sample estimate of variance. Given independent and identically distributed random variables $\mathrm{x}_1, \ldots, \mathrm{x}_N$, we might estimate the mean and the variance as

$$\hat{\mu} = \frac{1}{N} \sum\nolimits_i x_i, \quad \hat{\sigma}^2 = \frac{1}{N} \sum\nolimits_i (x_i - \hat{\mu})^2.$$

Such a mean estimate is unbiased, but the estimate of the variance is biased, because $\mathbb{E}[\hat{\sigma}^2] = (1 - \frac{1}{N})\sigma^2$; however, the bias of this estimate converges to zero for increasing N.

Also, an unbiased estimator does not necessarily have a small variance – in some cases, it can have a large variance, so a biased estimator with a smaller variance might be preferred.

Properties of Maximum Likelihood Estimation

Assume that the true data-generating distribution p_{data} lies within the model family $p_{\text{model}}(\bullet; \boldsymbol{\theta})$, and assume there exists a unique $\boldsymbol{\theta}_{p_{\text{data}}}$ such that $p_{\text{data}} = p_{\text{model}}(\bullet; \boldsymbol{\theta}_{p_{\text{data}}})$.

- MLE is a *consistent* estimator. If we denote $\boldsymbol{\theta}_m$ to be the parameters found by MLE for a training set with m examples generated by the data-generating distribution, then $\boldsymbol{\theta}_m$ converges in probability to $\boldsymbol{\theta}_{p_{\text{data}}}$.

  Formally, for any $\varepsilon > 0$, $P(\|\boldsymbol{\theta}_m - \boldsymbol{\theta}_{p_{\text{data}}}\| > \varepsilon) \to 0$ as $m \to \infty$.

- MLE is in a sense the *most statistically efficient*. For any consistent estimator, let us consider the average distance of $\boldsymbol{\theta}_m$ and $\boldsymbol{\theta}_{p_{\text{data}}}$: $\mathbb{E}_{\mathbf{x}_1,\ldots,\mathbf{x}_m \sim p_{\text{data}}}\left[\|\boldsymbol{\theta}_m - \boldsymbol{\theta}_{p_{\text{data}}}\|^2\right]$.
  It can be shown (Rao 1945, Cramér 1946) that no consistent estimator has lower mean squared error than the maximum likelihood estimator.

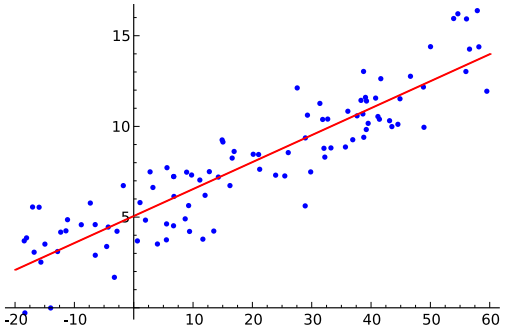
Therefore, for reasons of consistency and efficiency, maximum likelihood is often considered the preferred estimator for machine learning.

Loss Function

A model is usually trained in order to minimize the **loss** on the training data.

Assume our goal is to perform regression, i.e., to predict $p(y|\boldsymbol{x})$ for $y \in \mathbb{R}$. Let $\hat{y}(\boldsymbol{x};\boldsymbol{\theta})$ give a prediction of the mean of y.

We consider the most general distribution with the mean of the predicted value and an unknown but fixed variance σ^2 – the normal distribution. We define $p(y|\boldsymbol{x};\boldsymbol{\theta})$ as $\mathcal{N}(y; f(\boldsymbol{x};\boldsymbol{\theta}), \sigma^2)$ for some fixed σ^2.



https://upload.wikimedia.org/wikipedia/commons/3/3a

Then: $\arg\max_{\boldsymbol{\theta}} p(y|\mathbb{X};\boldsymbol{\theta}) = \arg\min_{\boldsymbol{\theta}} \frac{1}{N}\sum_{i=1}^{N}(\hat{y}(\boldsymbol{x}^{(i)};\boldsymbol{\theta}) - y^{(i)})^2$.

Based on the maximum likelihood principle, for estimation of parameters for *regression task*, the optimal computation of error (the loss) is the **mean square error**.

For estimation of parameters for *classification task*, based on the same principle, the optimal loss function is **negative log-likelihood** (**cross-entropy**), or **Kullback-Leibler divergence**.

Gradient Descent

Gradient Descent

Let $f(\boldsymbol{x};\boldsymbol{\theta})$ be a model with parameters $\boldsymbol{\theta}$. For a given per-example loss function L, denote

$$E(\boldsymbol{\theta}) = \mathbb{E}_{(\mathbf{x},\mathrm{y})\sim\hat{p}_{\text{data}}} L\big(f(\boldsymbol{x};\boldsymbol{\theta}), y\big).$$

Assuming we are minimizing a loss function

$$\arg\min_{\boldsymbol{\theta}} E(\boldsymbol{\theta}),$$

we may use *gradient descent*:

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \alpha \nabla_{\boldsymbol{\theta}} E(\boldsymbol{\theta}).$$

The constant α is called a **learning rate** and specifies the "length" of a step we perform in every iteration of the gradient descent.

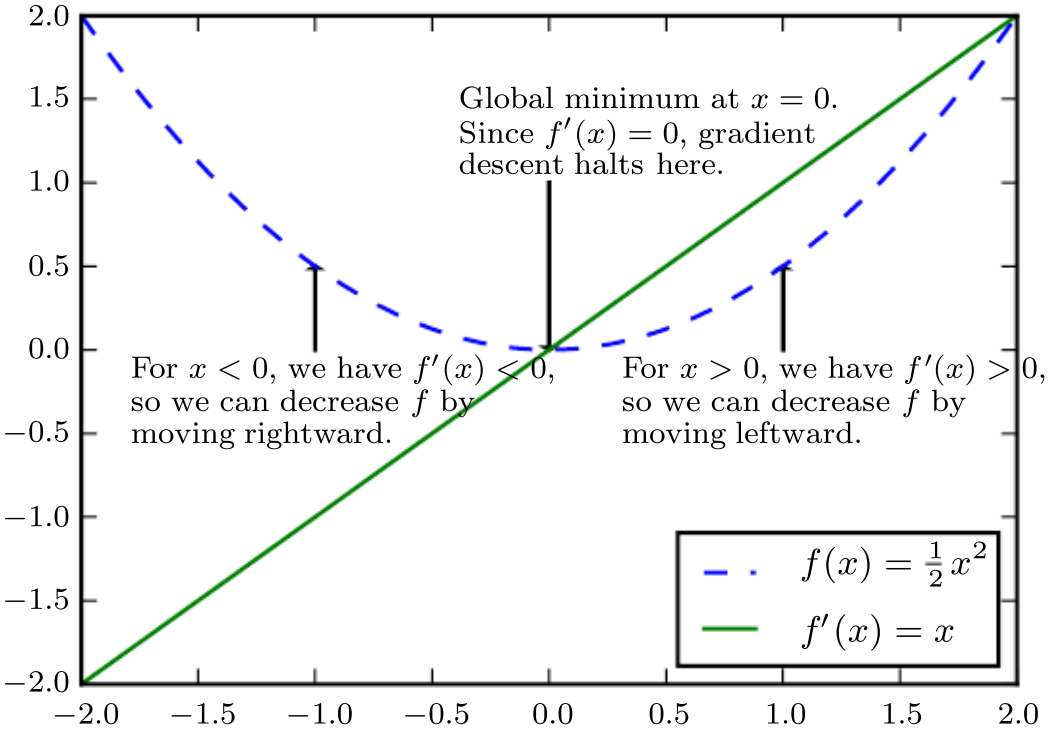


Figure 4.1 of "Deep Learning" book, https://www.deeplearningbook.org

Gradient Descent Variants

The gradient of the loss function $E(\boldsymbol{\theta})$ can be computed as

$$\nabla_{\boldsymbol{\theta}} E(\boldsymbol{\theta}) = \mathbb{E}_{(\mathbf{x},\mathrm{y})\sim\hat{p}_{\text{data}}} \nabla_{\boldsymbol{\theta}} L\big(f(\boldsymbol{x};\boldsymbol{\theta}), y\big).$$

- **(Standard/Batch) Gradient Descent**: We use all training data to compute $\nabla_{\boldsymbol{\theta}} E(\boldsymbol{\theta})$.
- **Stochastic (or Online) Gradient Descent**: We estimate $\nabla_{\boldsymbol{\theta}} E(\boldsymbol{\theta})$ using a single random example from the training data. Such an estimate is unbiased, but very noisy.

$$\nabla_{\boldsymbol{\theta}} E(\boldsymbol{\theta}) \approx \nabla_{\boldsymbol{\theta}} L\big(f(\boldsymbol{x};\boldsymbol{\theta}), y\big) \text{ for a randomly chosen } (\boldsymbol{x}, y) \text{ from } \hat{p}_{\text{data}}.$$

- **Minibatch SGD**: Trade-off between gradient descent and SGD – the expectation in $\nabla_{\boldsymbol{\theta}} E(\boldsymbol{\theta})$ is estimated using m random independent examples from the training data.

$$\nabla_{\boldsymbol{\theta}} E(\boldsymbol{\theta}) \approx \frac{1}{m} \sum_{i=1}^{m} \nabla_{\boldsymbol{\theta}} L\big(f(\boldsymbol{x}^{(i)};\boldsymbol{\theta}), y^{(i)}\big) \text{ for randomly chosen } (\boldsymbol{x}^{(i)}, y^{(i)}) \text{ from } \hat{p}_{\text{data}}.$$

Stochastic Gradient Descent Convergence

Assume that we perform a stochastic gradient descent, using a sequence of learning rates α_i, and using a noisy estimate $J(\boldsymbol{\theta})$ of the real gradient $\nabla_{\boldsymbol{\theta}} E(\boldsymbol{\theta})$:

$$\boldsymbol{\theta}_{i+1} \leftarrow \boldsymbol{\theta}_i - \alpha_i J(\boldsymbol{\theta}_i).$$

It can be proven (see Robbins and Monro algorithm, 1951) that if $|J(\boldsymbol{\theta})|$ is bounded and the loss function is strictly convex and continuous, then SGD converges to the unique optimum almost surely if the sequence of learning rates α_i fulfills the following conditions:

$$\sum_i \alpha_i = \infty, \quad \sum_i \alpha_i^2 < \infty.$$

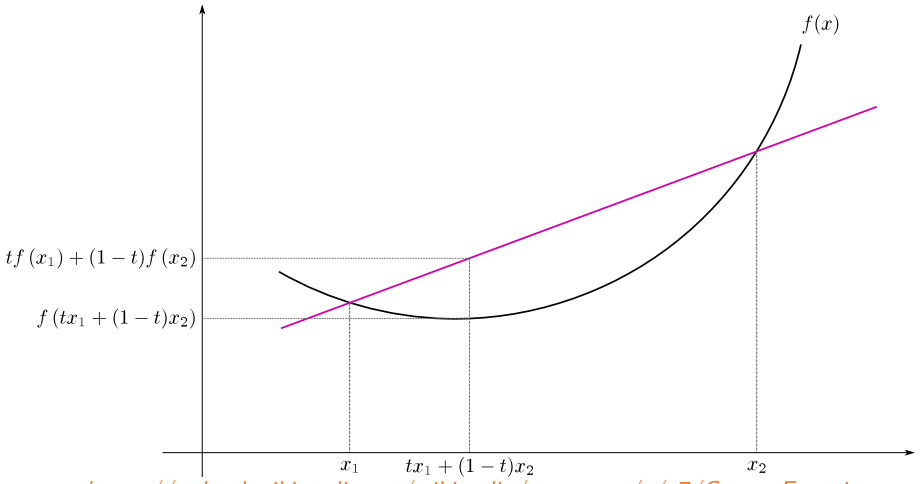
Note that the second condition implies that $\alpha_i \to 0$.

For nonconvex loss functions, we can get guarantees of converging to a *local* optimum only. However, note that finding the global minimum of even a boolean function is *at least NP-hard*.

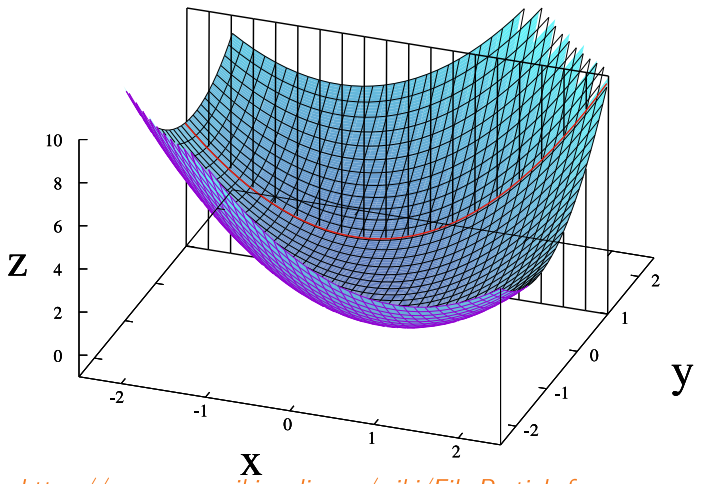
Stochastic Gradient Descent Convergence

Convex functions mentioned on the previous slide are such that for $\boldsymbol{u}, \boldsymbol{v}$ and real $0 \leq t \leq 1$,

$$f(t\boldsymbol{u} + (1-t)\boldsymbol{v}) \leq tf(\boldsymbol{u}) + (1-t)f(\boldsymbol{v}).$$



https://upload.wikimedia.org/wikipedia/commons/c/c7/ConvexFunction.svg



https://commons.wikimedia.org/wiki/File:Partial_func_eg.svg

A twice-differentiable function of a single variable is convex iff its second derivative is always nonnegative. (For functions of multiple variables, the Hessian must be positive semi-definite.)

A local minimum of a convex function is always a global minimum.

Well-known examples of convex functions are x^2, e^x, $-\log x$, MSE, σ+NLL, $\mathrm{softmax}$+NLL.

Loss Function Visualization

Visualization of loss function of ResNet-56 (0.85 million parameters) with/without skip connections:

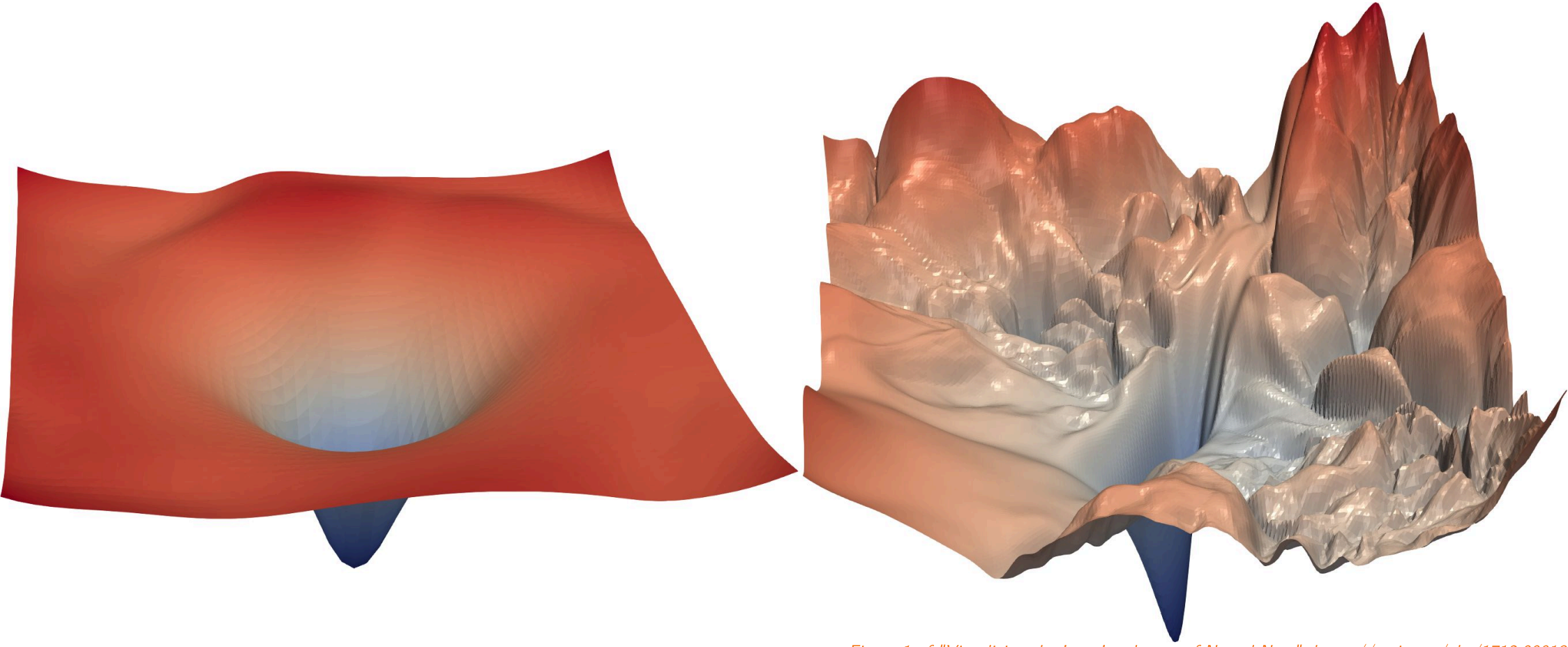
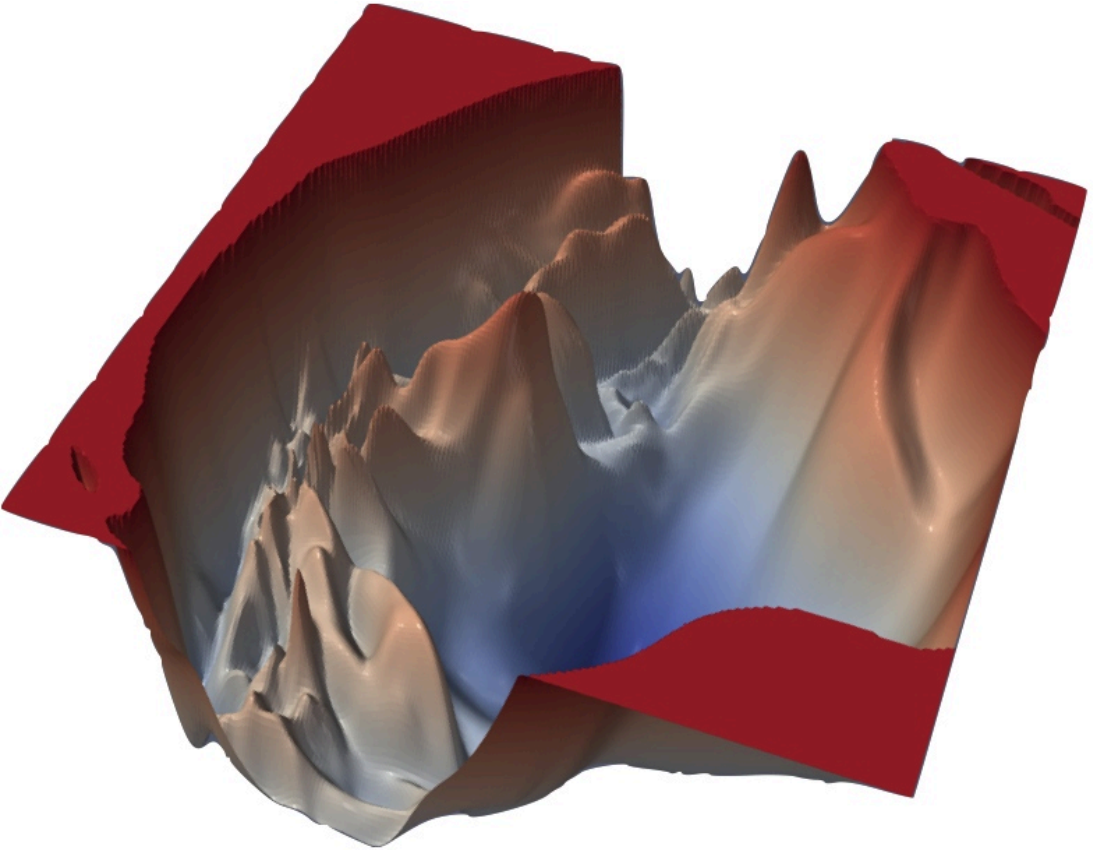


Figure 1 of "Visualizing the Loss Landscape of Neural Nets", https://arxiv.org/abs/1712.09913

Loss Function Visualization

Visualization of loss function of ResNet-110 without skip connections and DenseNet-121:



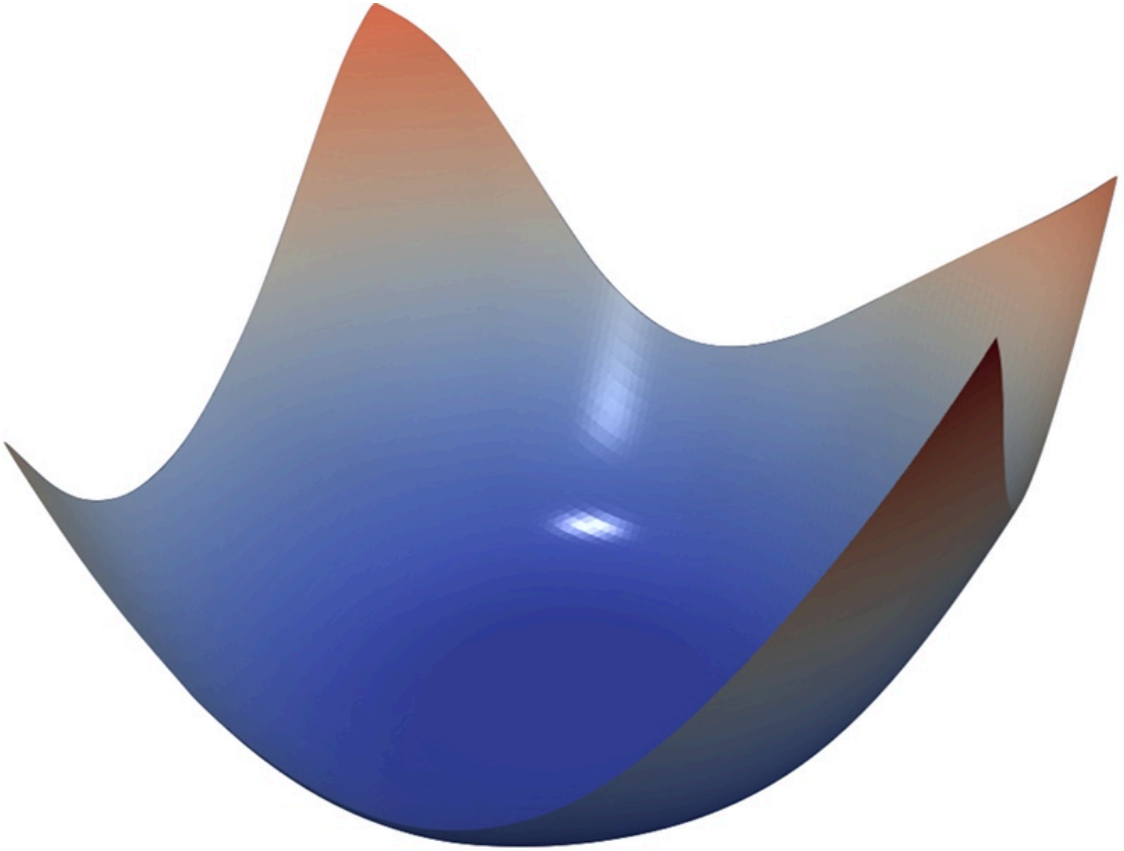
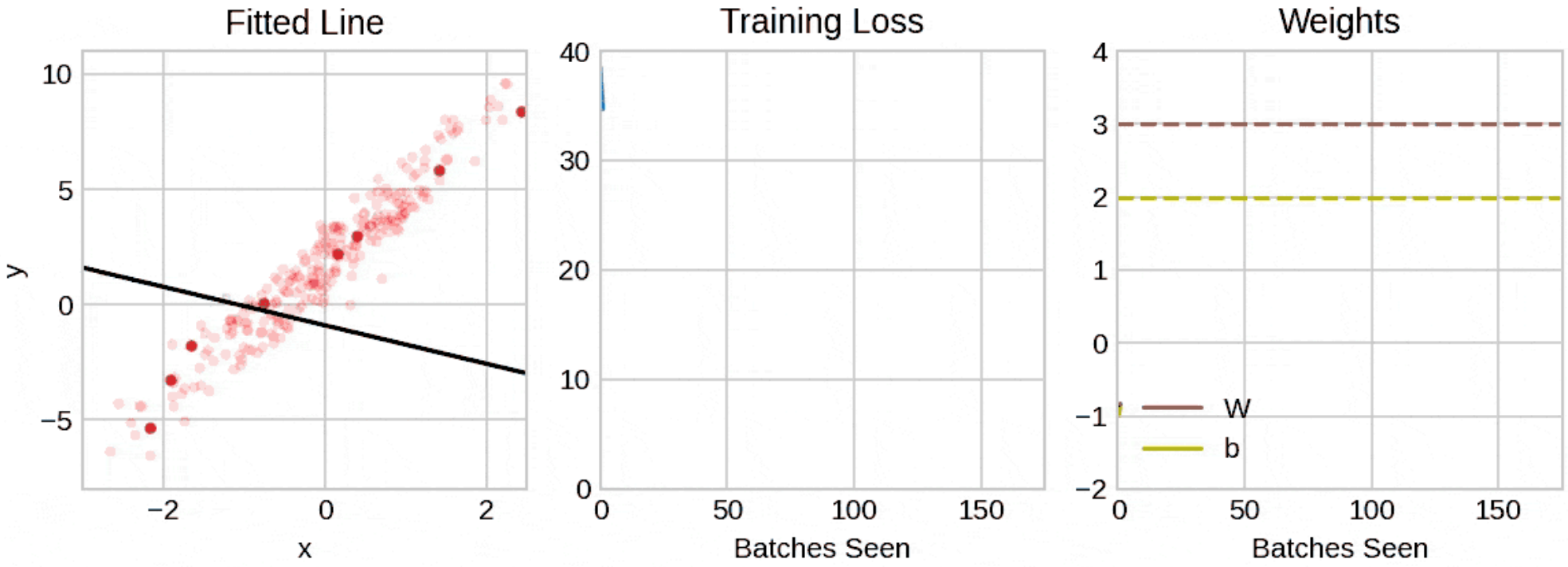


Figure 4 of "Visualizing the Loss Landscape of Neural Nets", https://arxiv.org/abs/1712.09913

Stochastic Gradient Descent Visualization

You can explore the interactive figures 2.1, 2.2, 2.4 at https://udlbook.github.io/udlfigures/.

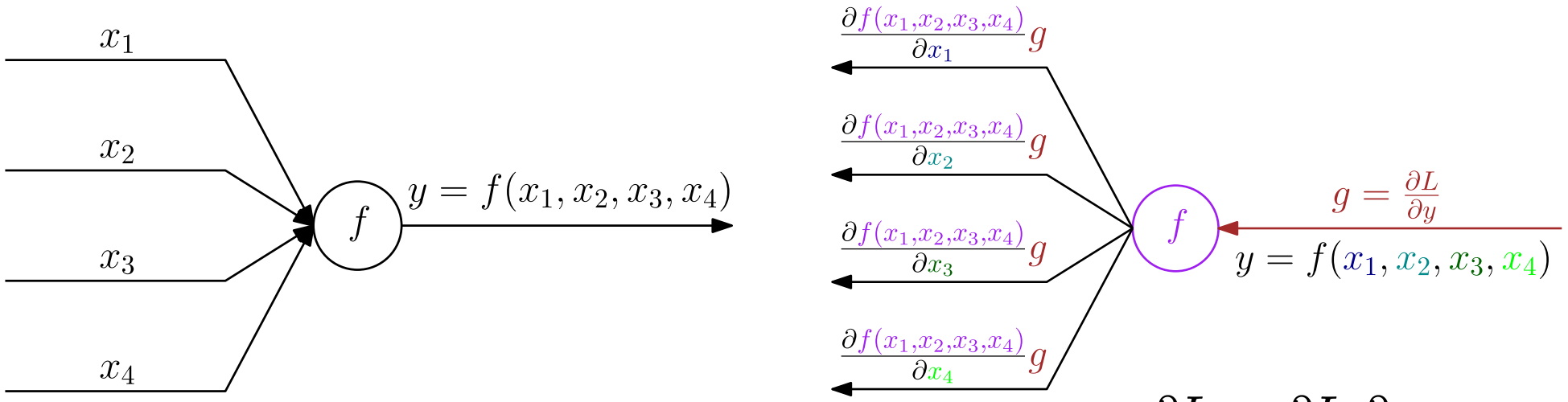


https://www.kaggle.com/code/ryanholbrook/stochastic-gradient-descent

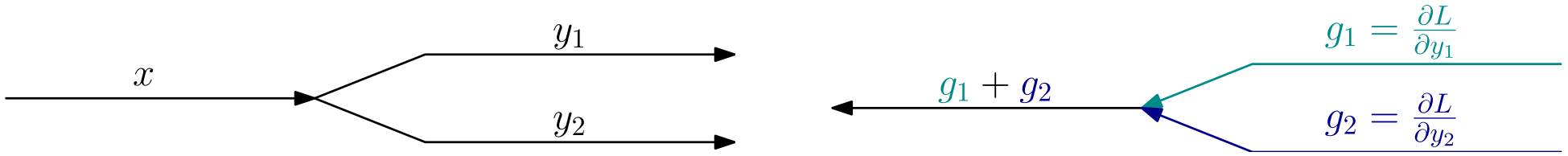
Backpropagation

Backpropagation

Assume we want to compute partial derivatives of a given loss function L.



The gradient computation is based on the chain rule of derivatives: $\dfrac{\partial L}{\partial x_i} = \dfrac{\partial L}{\partial y}\dfrac{\partial y}{\partial x_i}$.



Forward Propagation Algorithm

Forward Propagation

Input: Network with nodes $u^{(1)}, u^{(2)}, \ldots, u^{(n)}$ numbered in topological order.
Each node's value is computed as $u^{(i)} = f^{(i)}(A^{(i)})$ for $A^{(i)}$ being a set of values of the predecessors $P(u^{(i)})$ of $u^{(i)}$ (with indices $Pred(u^{(i)})$).

Output: Value of $u^{(n)}$.

- For $i = 1, \ldots, n$:
  - $\mathbb{A}^{(i)} \leftarrow \{u^{(j)} | j \in Pred(u^{(i)})\}$
  - $u^{(i)} \leftarrow f^{(i)}(\mathbb{A}^{(i)})$
- Return $u^{(n)}$

Simple Variant of Backpropagation

Input: The network as in the Forward propagation algorithm.
Output: Partial derivatives $g^{(i)} = \frac{\partial u^{(n)}}{\partial u^{(i)}}$ of $u^{(n)}$ with respect to all $u^{(i)}$.

- Run forward propagation to compute all $u^{(i)}$
- $g^{(n)} = 1$
- For $i = n-1, \ldots, 1$:
  - $g^{(i)} \leftarrow \sum_{j:i \in Pred(u^{(j)})} g^{(j)} \frac{\partial u^{(j)}}{\partial u^{(i)}}$
- Return $\left(g^{(1)}, g^{(2)}, \ldots, g^{(n)}\right)$

In practice, we do not usually represent networks as collections of scalar nodes; instead we represent them as collections of tensor functions – most usually functions $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m$. Then $\frac{\partial f(\boldsymbol{x})}{\partial \boldsymbol{x}}$ is a Jacobian matrix. However, the backpropagation algorithm is analogous.

Hidden Layers Derivatives

- σ:

$$\frac{\partial \sigma(x)}{\partial x} = \sigma(x) \cdot \big(1 - \sigma(x)\big)$$

- $\tanh$:

$$\frac{\partial \tanh(x)}{\partial x} = 1 - \tanh(x)^2$$

- ReLU:

$$\frac{\partial \operatorname{ReLU}(x)}{\partial x} = \left\{ \begin{matrix} 1 & \text{if } x > 0 \\ \text{NaN} & \text{if } x = 0 \\ 0 & \text{if } x < 0 \end{matrix} \right\} \xlongequal{\text{assuming } \frac{\partial \operatorname{ReLU}(x)}{\partial x}(0)=0} \big[x > 0\big] = \big[\operatorname{ReLU}(x) > 0\big]$$

Stochastic Gradient Descents

Stochastic Gradient Descent

Stochastic Gradient Descent (SGD) Algorithm

Input: NN computing function $f(\boldsymbol{x};\boldsymbol{\theta})$ with initial value of parameters $\boldsymbol{\theta}$.

Input: Learning rate α.

Output: Updated parameters $\boldsymbol{\theta}$.

- Repeat until stopping criterion is met:
  - Sample a minibatch of m training examples $(\boldsymbol{x}^{(i)}, y^{(i)})$
    - in theory, we could sample each minibatch independently;
    - however, almost every time we want to process all training instances before repeating them, which can be implemented by generating a random permutation and then splitting it into minibatch-sized chunks
      - one pass through the data is called an **epoch**
  - $\boldsymbol{g} \leftarrow \frac{1}{m}\sum_i \nabla_{\boldsymbol{\theta}} L\big(f(\boldsymbol{x}^{(i)};\boldsymbol{\theta}), y^{(i)}\big)$
  - $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \alpha \boldsymbol{g}$

SGD With Momentum

SGD With Momentum

Input: NN computing function $f(\boldsymbol{x};\boldsymbol{\theta})$ with initial value of parameters $\boldsymbol{\theta}$.
Input: Learning rate α, momentum β.
Output: Updated parameters $\boldsymbol{\theta}$.

- $\boldsymbol{v} \leftarrow \boldsymbol{0}$
- Repeat until stopping criterion is met:
  - Sample a minibatch of m training examples $(\boldsymbol{x}^{(i)}, y^{(i)})$
  - $\boldsymbol{g} \leftarrow \frac{1}{m}\sum_i \nabla_{\boldsymbol{\theta}} L\big(f(\boldsymbol{x}^{(i)};\boldsymbol{\theta}), y^{(i)}\big)$
  - $\boldsymbol{v} \leftarrow \beta\boldsymbol{v} + \boldsymbol{g}$
  - $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \alpha\boldsymbol{v}$

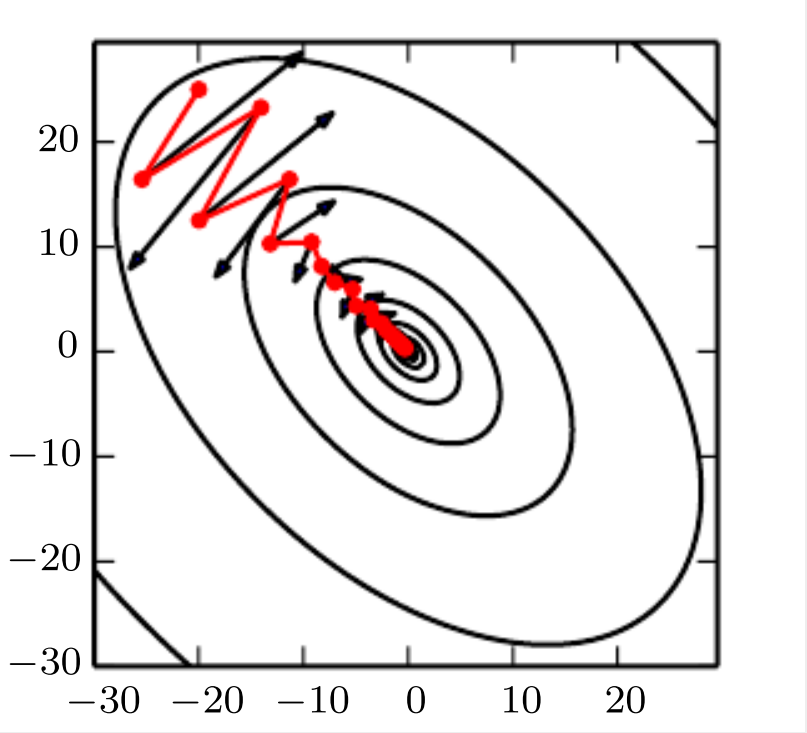
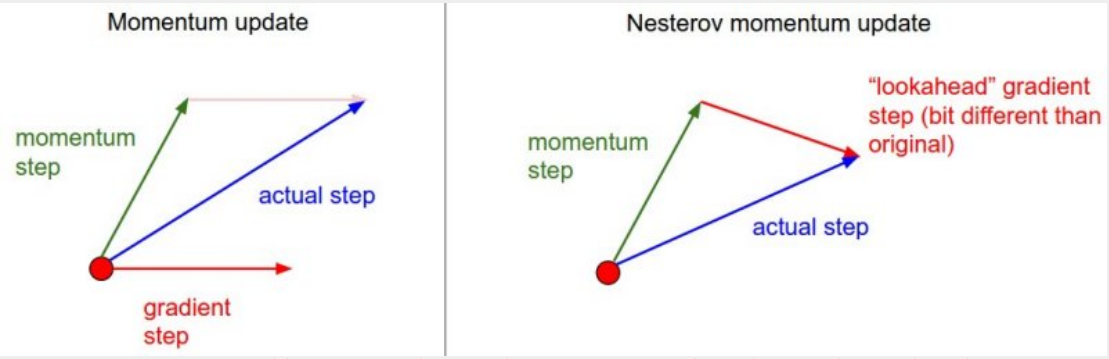


Figure 8.5 of "Deep Learning" book, https://www.deeplearningbook.org

A nice writeup about momentum can be found on https://distill.pub/2017/momentum/.

SGD With Nesterov Momentum

Input: NN computing function $f(\boldsymbol{x};\boldsymbol{\theta})$ with initial value of parameters $\boldsymbol{\theta}$.
Input: Learning rate α, momentum β.
Output: Updated parameters $\boldsymbol{\theta}$.



https://github.com/cs231n/cs231n.github.io/blob/master/assets/nn3/nesterov.jpeg

- $\boldsymbol{v} \leftarrow \boldsymbol{0}$
- Repeat until stopping criterion is met:
  - Sample a minibatch of m training examples $(\boldsymbol{x}^{(i)}, y^{(i)})$
    # $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \alpha\beta\boldsymbol{v} - \alpha\boldsymbol{g}$ is decomposed to two steps:
  - $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \alpha\beta\boldsymbol{v}$
  - $\boldsymbol{g} \leftarrow \frac{1}{m}\sum_i \nabla_{\boldsymbol{\theta}} L\big(f(\boldsymbol{x}^{(i)};\boldsymbol{\theta}), y^{(i)}\big)$
  - $\boldsymbol{v} \leftarrow \beta\boldsymbol{v} + \boldsymbol{g}$
  - $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \alpha\boldsymbol{g}$

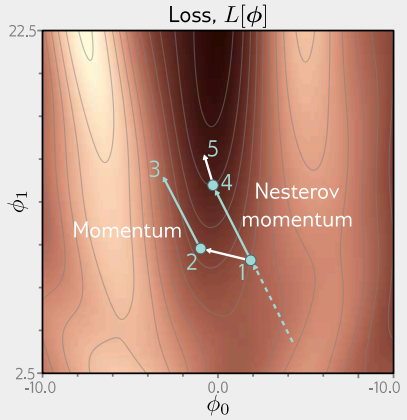


Figure 6.8 of "Understanding Deep Learning", https://udlbook.github.io/udlbook/

Algorithms with Adaptive Learning Rates

Algorithms with Adaptive Learning Rates

AdaGrad (2011)

Input: NN computing function $f(\boldsymbol{x};\boldsymbol{\theta})$ with initial value of parameters $\boldsymbol{\theta}$.
Input: Learning rate α, constant ε (usually 10^{-7}).
Output: Updated parameters $\boldsymbol{\theta}$.

- $\boldsymbol{r} \leftarrow \mathbf{0}$
- Repeat until stopping criterion is met:
  - Sample a minibatch of m training examples $(\boldsymbol{x}^{(i)}, y^{(i)})$
  - $\boldsymbol{g} \leftarrow \frac{1}{m}\sum_i \nabla_{\boldsymbol{\theta}} L(f(\boldsymbol{x}^{(i)};\boldsymbol{\theta}), y^{(i)})$
  - $\boldsymbol{r} \leftarrow \boldsymbol{r} + \boldsymbol{g}^2$
  - $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \frac{\alpha}{\sqrt{\boldsymbol{r}}+\varepsilon}\boldsymbol{g}$

- The $\boldsymbol{g}^2$ and $\frac{\alpha}{\sqrt{\boldsymbol{r}}+\varepsilon}\boldsymbol{g}$ are computed element-wise, i.e., $\boldsymbol{g}^2 = \boldsymbol{g} \odot \boldsymbol{g}$. It might be better to write $\frac{\alpha}{\sqrt{\boldsymbol{r}}+\varepsilon} \odot \boldsymbol{g}$, but it is not done in the papers, so we are keeping the usual notation.

Algorithms with Adaptive Learning Rates

AdaGrad has favourable convergence properties (being faster than regular SGD) for convex loss landscapes. In this settings, gradients converge to zero reasonably fast.

However, for nonconvex losses, gradients can stay quite large for a long time. In that case, the algorithm behaves as if decreasing learning rate by a factor of $1/\sqrt{t}$, because if each

$$\boldsymbol{g} \approx \boldsymbol{g}_0,$$

then after t steps

$$\boldsymbol{r} \approx t \cdot \boldsymbol{g}_0^2,$$

and therefore

$$\frac{\alpha}{\sqrt{\boldsymbol{r}} + \varepsilon} \approx \frac{\alpha/\sqrt{t}}{\sqrt{\boldsymbol{g}_0^2} + \varepsilon/\sqrt{t}}.$$

Algorithms with Adaptive Learning Rates

RMSProp (2012)

Input: NN computing function $f(\boldsymbol{x};\boldsymbol{\theta})$ with initial value of parameters $\boldsymbol{\theta}$.
Input: Learning rate α, momentum β (usually 0.9), constant ε (usually 10^{-7}).
Output: Updated parameters $\boldsymbol{\theta}$.

- $\boldsymbol{r} \leftarrow \mathbf{0}$
- Repeat until stopping criterion is met:
  - Sample a minibatch of m training examples $(\boldsymbol{x}^{(i)}, y^{(i)})$
  - $\boldsymbol{g} \leftarrow \frac{1}{m}\sum_i \nabla_{\boldsymbol{\theta}} L(f(\boldsymbol{x}^{(i)};\boldsymbol{\theta}), y^{(i)})$
  - $\boldsymbol{r} \leftarrow \beta\boldsymbol{r} + (1-\beta)\boldsymbol{g}^2$
  - $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \frac{\alpha}{\sqrt{\boldsymbol{r}}+\varepsilon}\boldsymbol{g}$

However, after first step, $\boldsymbol{r} = (1-\beta)\boldsymbol{g}^2$, which for default $\beta = 0.9$ is

$$\boldsymbol{r} = 0.1\boldsymbol{g}^2,$$

so $\boldsymbol{r}$ is a biased estimate of $\mathbb{E}[\boldsymbol{g}^2]$ (but the bias converges to zero exponentially fast).

Algorithms with Adaptive Learning Rates

After t steps, we have

$$\boldsymbol{r}_t = (1-\beta)\sum_{i=1}^{t}\beta^{t-i}\boldsymbol{g}_i^2.$$

Therefore, we are computing weighted average of elements with a total weight of

$$(1-\beta)\sum_{i=1}^{t}\beta^{t-i} = (1-\beta)\frac{1-\beta^t}{1-\beta} = 1-\beta^t.$$

In other words,

$$\mathbb{E}[\boldsymbol{r}_t] \approx \mathbb{E}[\boldsymbol{g}^2]\cdot(1-\beta^t),$$

so in order to obtain an unbiased estimate, we need to divide $\boldsymbol{r}_t$ by $(1-\beta^t)$, and analogously for the correction of the first power of $\boldsymbol{g}$.

Algorithms with Adaptive Learning Rates

Adam (2014)

Input: NN computing function $f(\boldsymbol{x};\boldsymbol{\theta})$ with initial value of parameters $\boldsymbol{\theta}$.
Input: Learning rate α (default 0.001), constant ε (usually 10^{-7}).
Input: Momentum β_1 (default 0.9), momentum β_2 (default 0.999).
Output: Updated parameters $\boldsymbol{\theta}$.

- $\boldsymbol{s} \leftarrow \mathbf{0}$, $\boldsymbol{r} \leftarrow \mathbf{0}$, $t \leftarrow 0$
- Repeat until stopping criterion is met:
  - Sample a minibatch of m training examples $(\boldsymbol{x}^{(i)}, y^{(i)})$
  - $\boldsymbol{g} \leftarrow \frac{1}{m}\sum_i \nabla_{\boldsymbol{\theta}} L(f(\boldsymbol{x}^{(i)};\boldsymbol{\theta}), y^{(i)})$
  - $t \leftarrow t+1$
  - $\boldsymbol{s} \leftarrow \beta_1 \boldsymbol{s} + (1-\beta_1)\boldsymbol{g}$ *(biased first moment estimate)*
  - $\boldsymbol{r} \leftarrow \beta_2 \boldsymbol{r} + (1-\beta_2)\boldsymbol{g}^2$ *(biased second moment estimate)*
  - $\hat{\boldsymbol{s}} \leftarrow \boldsymbol{s}/(1-\beta_1^t)$, $\hat{\boldsymbol{r}} \leftarrow \boldsymbol{r}/(1-\beta_2^t)$ *(unbiased estimates of the moments)*
  - $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \frac{\alpha}{\sqrt{\hat{\boldsymbol{r}}}+\varepsilon}\hat{\boldsymbol{s}}$

Visualization of Gradient Normalization Effect

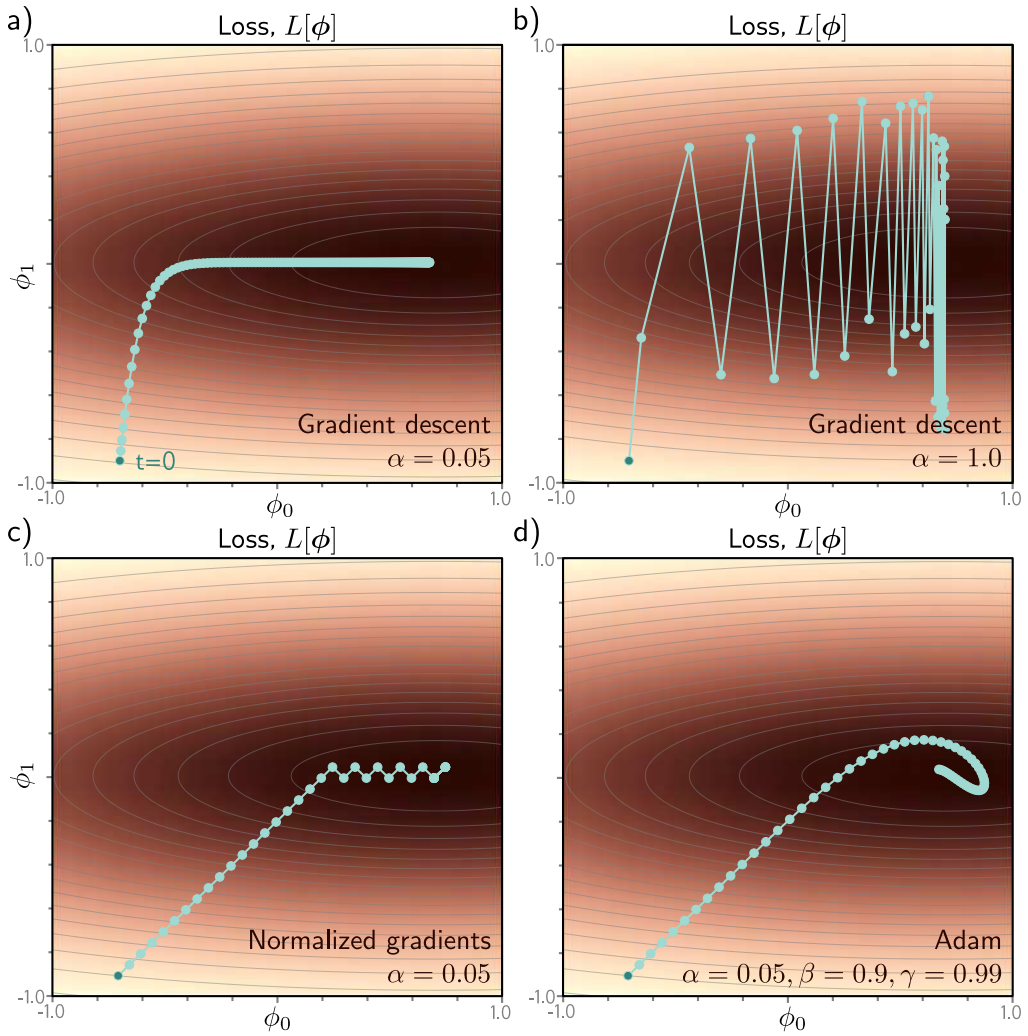
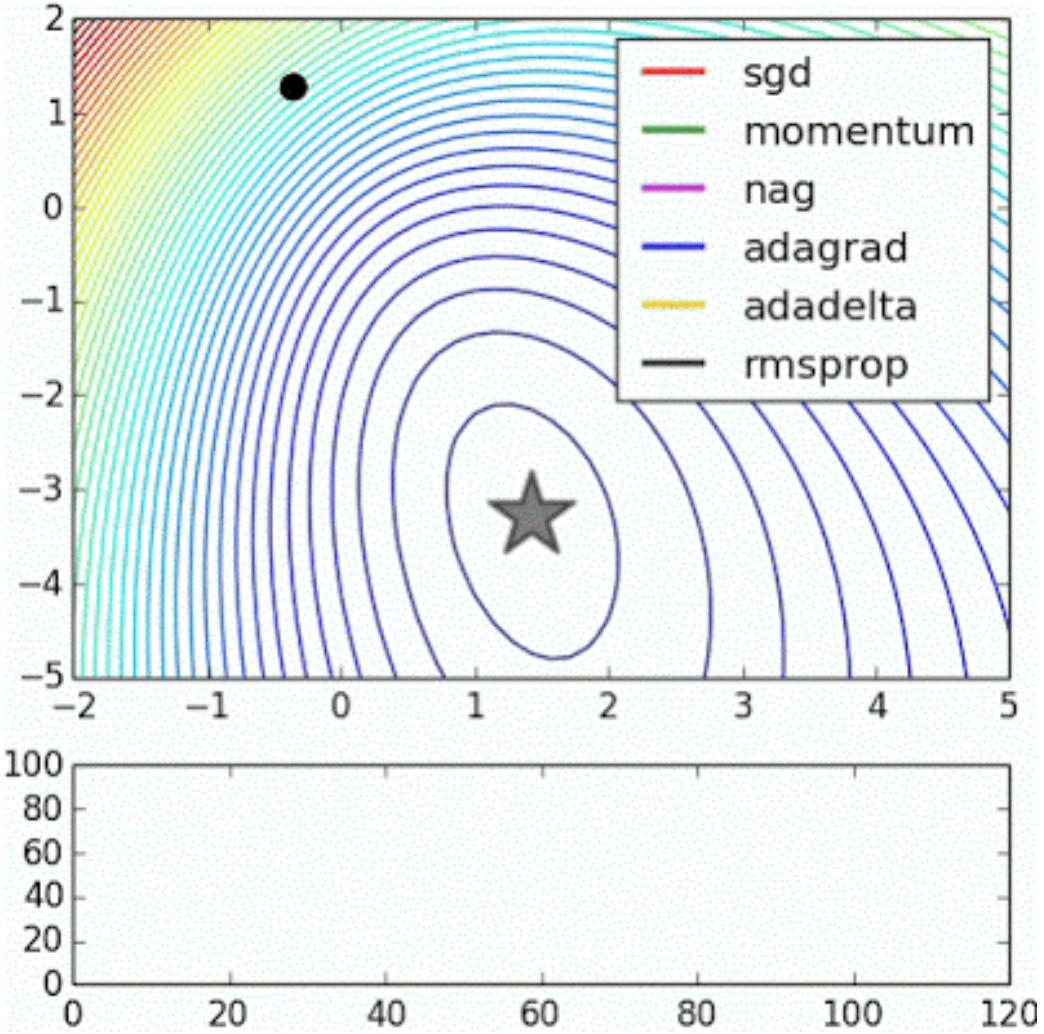


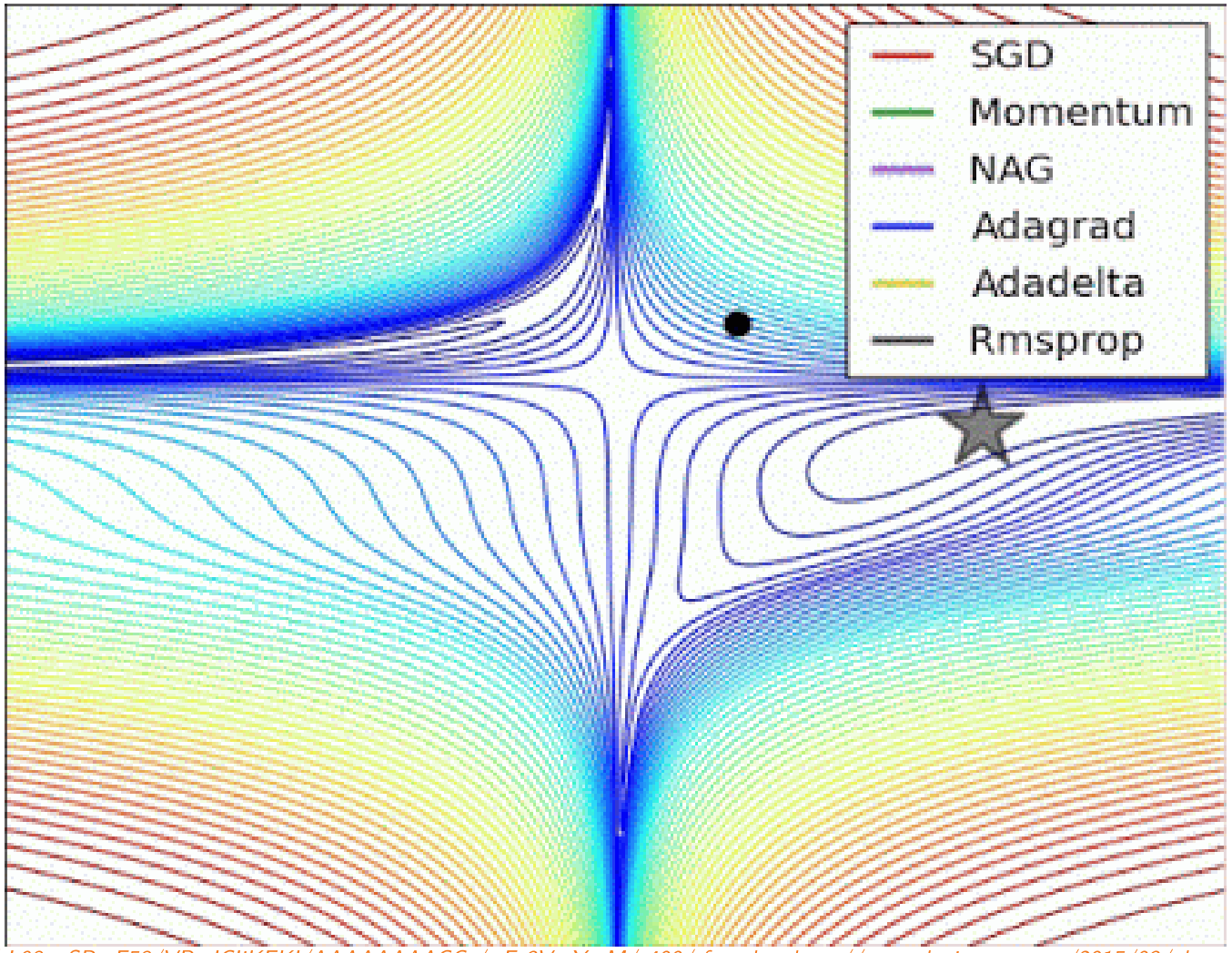
Figure 6.9 of "Understanding Deep Learning", https://udlbook.github.io/udlbook/

Adaptive Optimizers Animations



http://2.bp.blogspot.com/-q6l20Vs4P_w/VPmIC7sEhnI/AAAAAAAACC4/g3UOUX2r_yA/s400/ found at http://www.denizyuret.com/2015/03/alec-radfords-animations-for.html

Adaptive Optimizers Animations



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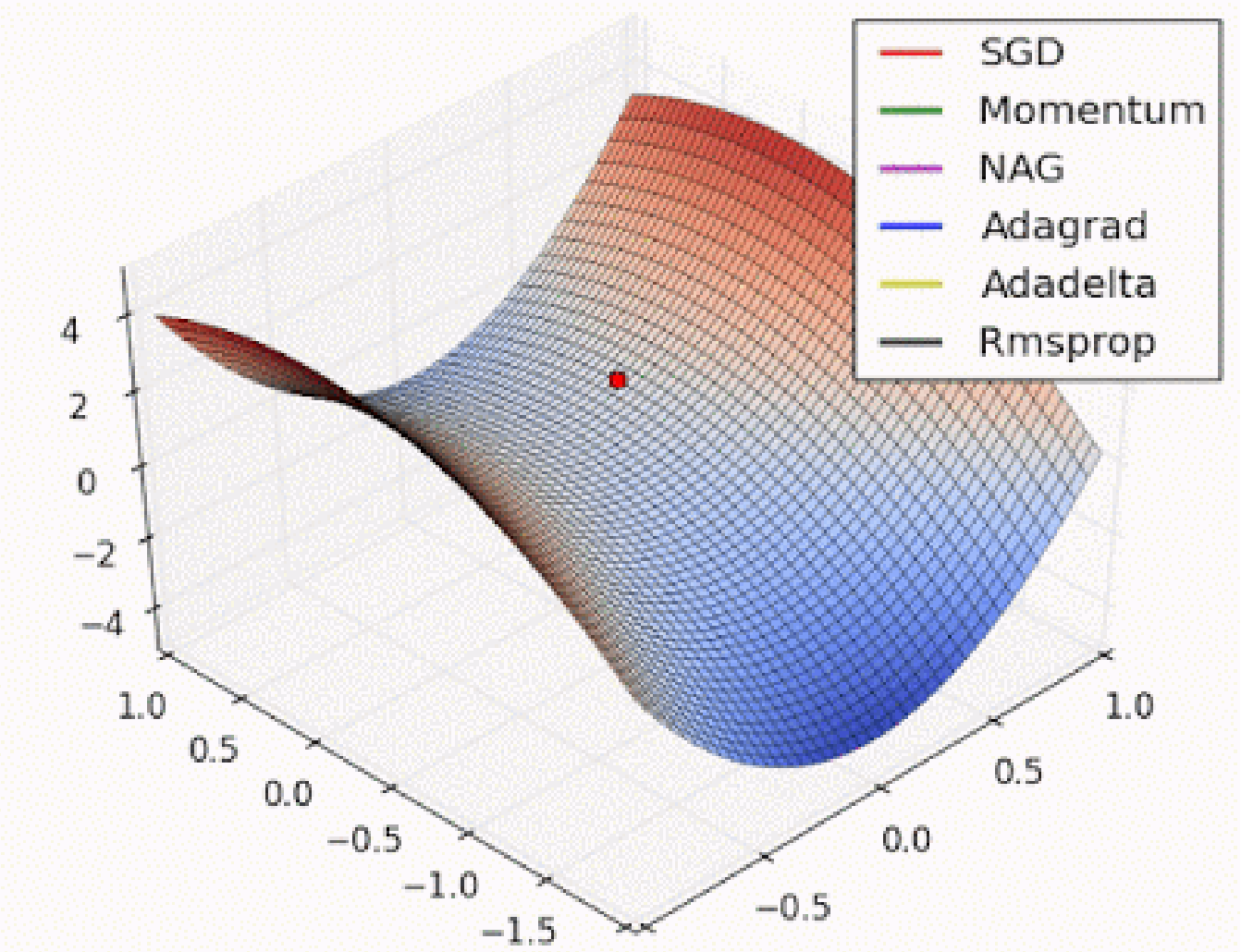
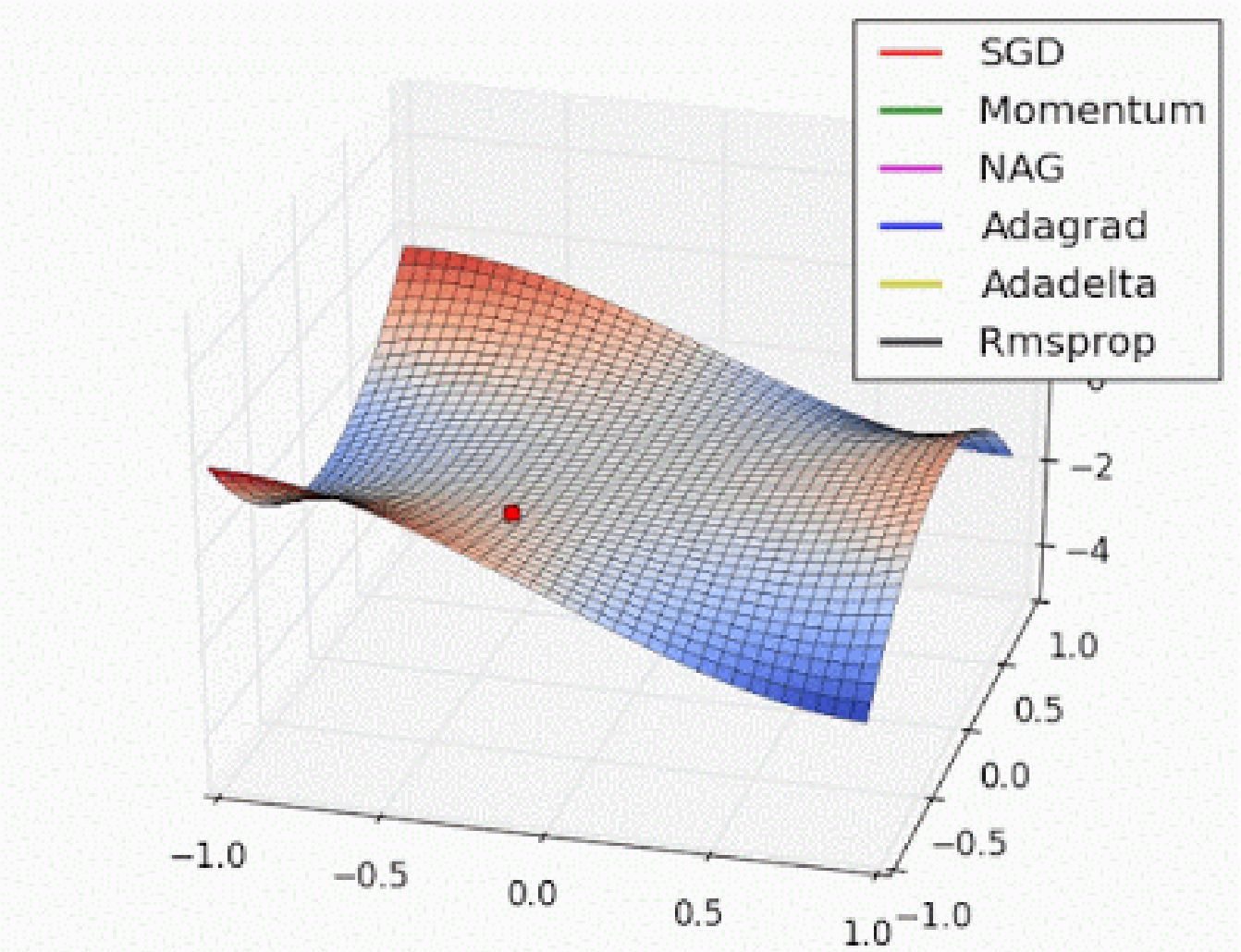
Adaptive Optimizers Animations



http://3.bp.blogspot.com/-nrtJPrdBWuE/VPmIB46F2aI/AAAAAAAACCw/vaE_B0SVy5k/s400/ found at http://www.denizyuret.com/2015/03/alec-radfords-animations-for.html

Adaptive Optimizers Animations



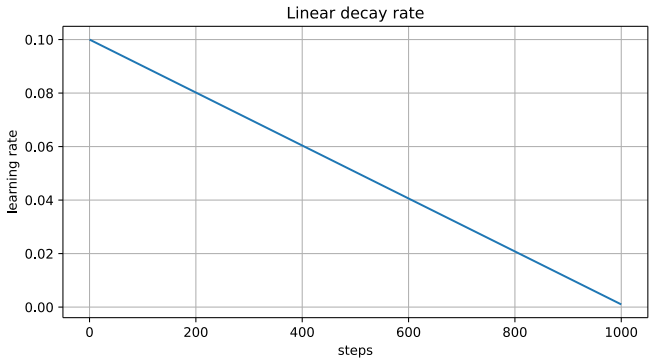
http://1.bp.blogspot.com/-K_X-yud8nj8/VPmIBxwGlsI/AAAAAAAACC0/JS-h1fa09EQ/s400/ found at http://www.denizyuret.com/2015/03/alec-radfords-animations-for.html

Learning Rate Schedules

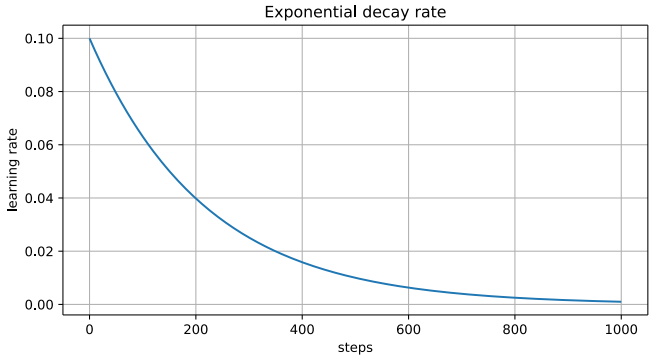
Learning Rate Schedules

Even if RMSProp and Adam are adaptive, they still usually require carefully tuned decreasing learning rate for top-notch performance.

- **Polynomial decay**: learning rate is multiplied by some polynomial of the current update number t.
  - **Linear decay** uses $\alpha_t = \alpha_{\text{initial}} \cdot \left(1 - \frac{t}{\text{max steps}}\right)$ and has theoretical guarantees of convergence, but is usually too fast for deep neural networks.
  - **Inverse square root decay** uses $\alpha_t = \alpha_{\text{initial}} \cdot \frac{1}{\sqrt{t}}$ and is used by advanced machine translation models.



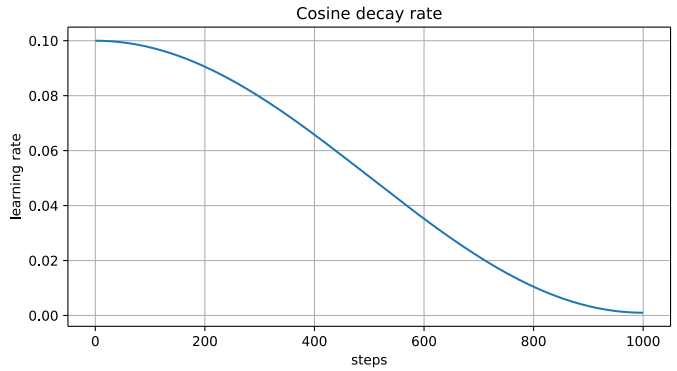
- **Exponential decay**: learning rate is multiplied by a constant each minibatch/epoch/several epochs.
  - $\alpha_t = \alpha_{\text{initial}} \cdot c^t$
  - Often used for convolutional networks (image recognition etc.).



Learning Rate Schedules

- **Cosine decay**: The cosine decay has became quite popular in the past years, both for training and finetuning.

$$\alpha_t = \alpha_{\text{initial}} \cdot \frac{1}{2}\left(1 + \cos\left(\pi \cdot \frac{t}{\text{max steps}}\right)\right)$$
$$= \alpha_{\text{initial}} \cdot \cos^2\left(\frac{\pi}{2} \cdot \frac{t}{\text{max steps}}\right)$$



- Cyclic restarts, warmup, ...

Learning Rate Schedules

The `torch.optim.lr_scheduler` offers several learning rate schedulers, which can be used directly for learning rate management.

- `torch.optim.lr_scheduler.LinearLR`
- `torch.optim.lr_scheduler.PolynomialLR`
- `torch.optim.lr_scheduler.ExponentialLR`
- `torch.optim.lr_scheduler.StepLR`
- `torch.optim.lr_scheduler.MultiStepLR`
- `torch.optim.lr_scheduler.CosineAnnealingLR`
- `torch.optim.lr_scheduler.ChainedScheduler`
- `torch.optim.lr_scheduler.LambdaLR`
- ...

Why do Neural Networks Generalize so Well: Double Descent

Why do Neural Networks Generalize so Well: Double Descent

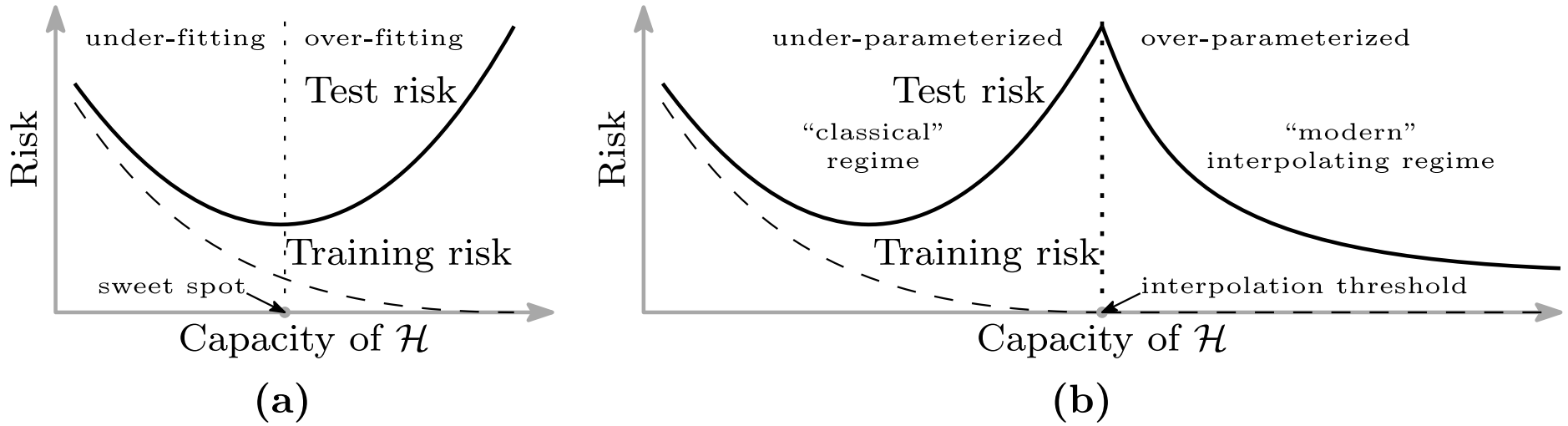


Figure 1: **Curves for training risk (dashed line) and test risk (solid line).** (**a**) The classical *U-shaped risk curve* arising from the bias-variance trade-off. (**b**) The *double descent risk curve*, which incorporates the U-shaped risk curve (i.e., the "classical" regime) together with the observed behavior from using high capacity function classes (i.e., the "modern" interpolating regime), separated by the interpolation threshold. The predictors to the right of the interpolation threshold have zero training risk.

Figure 1 of "Reconciling modern machine learning practice and the bias-variance trade-off", https://arxiv.org/abs/1812.11118

Deep Double Descent

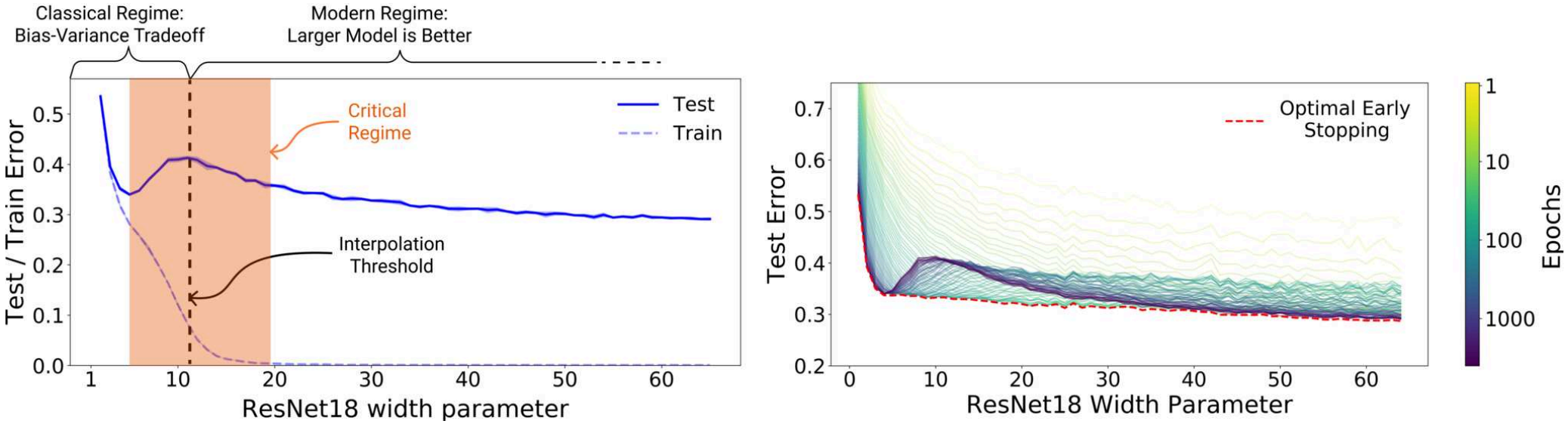


Figure 1: **Left:** Train and test error as a function of model size, for ResNet18s of varying width on CIFAR-10 with 15% label noise. **Right:** Test error, shown for varying train epochs. All models trained using Adam for 4K epochs. The largest model (width 64) corresponds to standard ResNet18.

Figure 1 of "Deep Double Descent: Where Bigger Models and More Data Hurt", https://arxiv.org/abs/1912.02292

Deep Double Descent – Effective Model Complexity

The authors define the **Effective Model Complexity** (**EMC**) of a training procedure $\mathcal{T}$ with respect to distribution $\mathcal{D}$ and parameter $\varepsilon > 0$ as

$$\mathrm{EMC}_{\mathcal{D},\varepsilon}(\mathcal{T}) \stackrel{\text{def}}{=} \max\left\{n \,\middle|\, \mathbb{E}_{S\sim\mathcal{D}^n}[\mathrm{Error}_S(\mathcal{T}(S))] \leq \varepsilon\right\},$$

where $\mathrm{Error}_S(M)$ is the mean error of a model M on the train samples S.

Hypothesis: For any natural data distribution $\mathcal{D}$, neural-network-based training procedure $\mathcal{T}$, and small $\varepsilon > 0$, if we consider the task of predicting labels based on n samples from $\mathcal{D}$, then:

- **Under-parametrized regime.** If $\mathrm{EMC}_{\mathcal{D},\varepsilon}(\mathcal{T})$ is sufficiently smaller than n, any perturbation of $\mathcal{T}$ that increases its effective complexity will decrease the test error.
- **Critically parametrized regime.** If $\mathrm{EMC}_{\mathcal{D},\varepsilon}(\mathcal{T}) \approx n$, then a perturbation of $\mathcal{T}$ that increases its effective complexity might decrease **or increase** the test error.
- **Over-parametrized regime.** If $\mathrm{EMC}_{\mathcal{D},\varepsilon}(\mathcal{T})$ is sufficiently larger than n, any perturbation of $\mathcal{T}$ that increases its effective complexity will decrease the test error.

Why do Neural Networks Generalize so Well: Double Descent

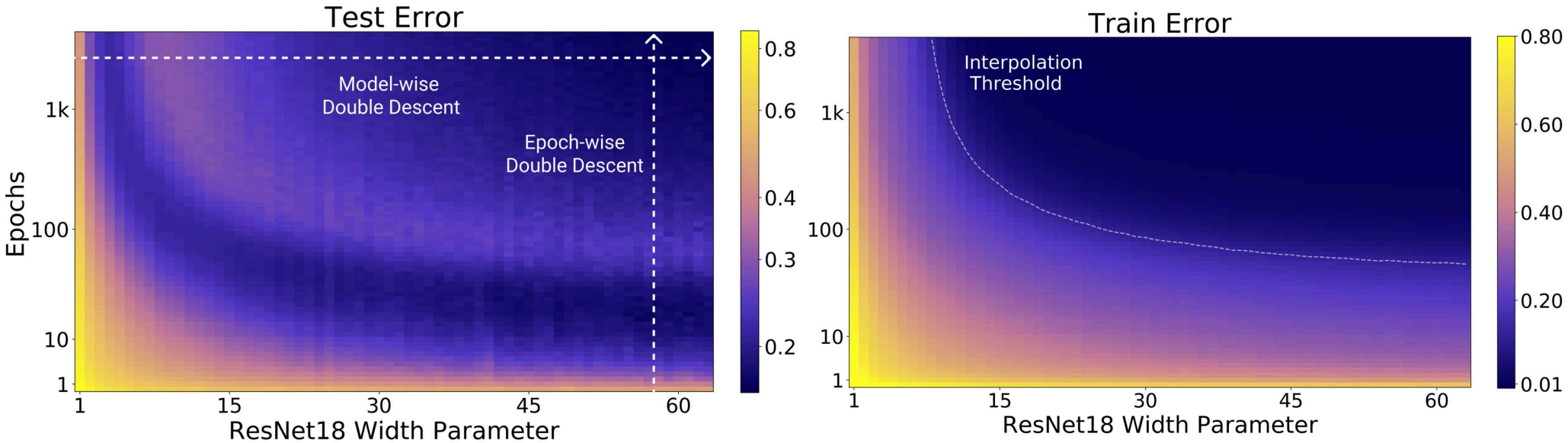
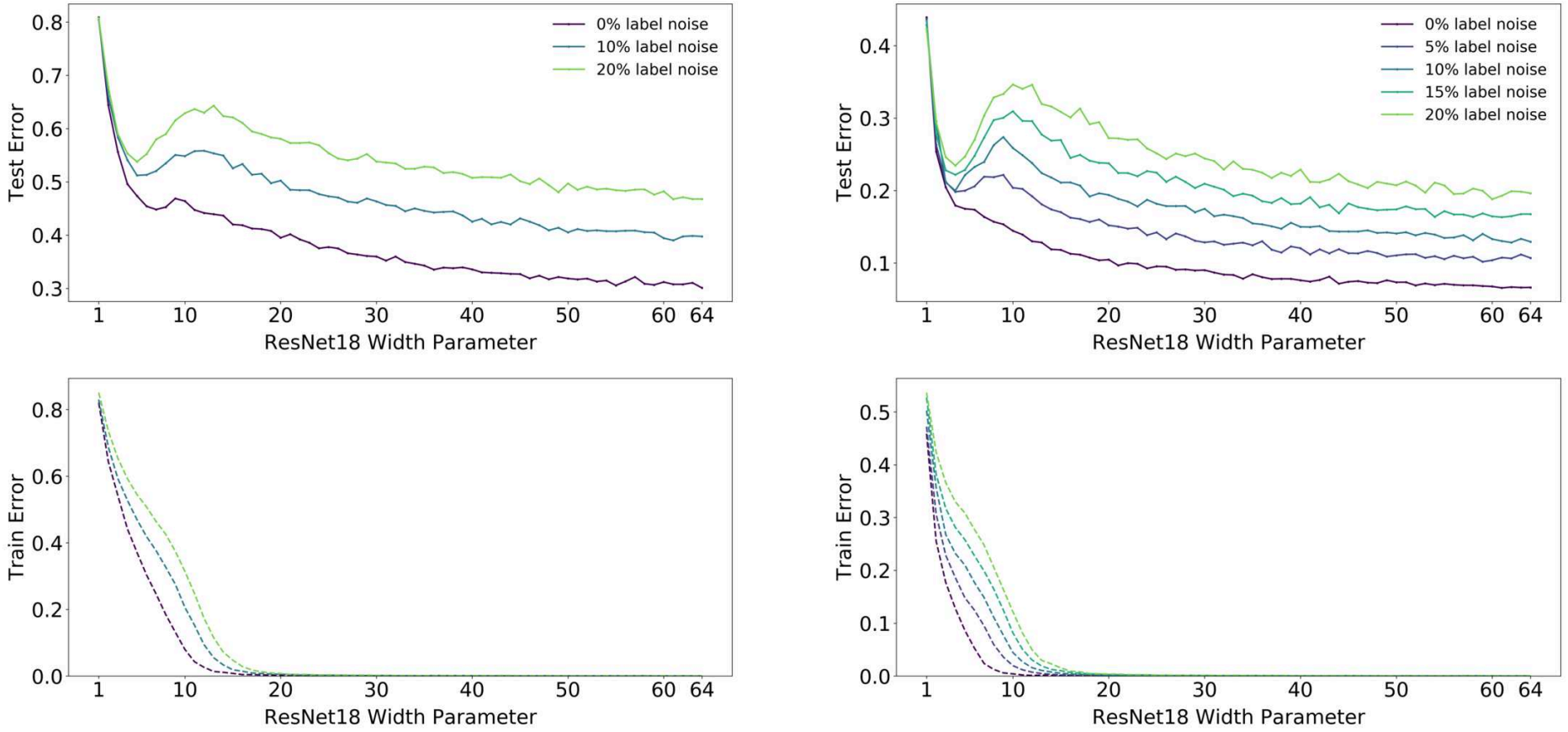


Figure 2: **Left:** Test error as a function of model size and train epochs. The horizontal line corresponds to model-wise double descent–varying model size while training for as long as possible. The vertical line corresponds to epoch-wise double descent, with test error undergoing double-descent as train time increases. **Right** Train error of the corresponding models. All models are Resnet18s trained on CIFAR-10 with 15% label noise, data-augmentation, and Adam for up to 4K epochs.

Figure 2 of "Deep Double Descent: Where Bigger Models and More Data Hurt", https://arxiv.org/abs/1912.02292

Why do Neural Networks Generalize so Well: Double Descent



(a) **CIFAR-100.** There is a peak in test error even with no label noise.

(b) **CIFAR-10.** There is a "plateau" in test error around the interpolation point with no label noise, which develops into a peak for added label noise.

Figure 4 of "Deep Double Descent: Where Bigger Models and More Data Hurt", https://arxiv.org/abs/1912.02292

Main Takeaways

- We train neural networks by minimizing **training error** on the training data.
- We measure the performance of a model by evaluating **generalization error** on unseen examples (usually a fixed test set).

The training process is based on several components:

- **Maximum likelihood estimation** provides a loss for any output distribution. This loss is in some sense the most statistically efficient.
- **Gradient descent** is an algorithm for performing the training itself, based on the (stochastic) gradient of the loss function.
- **Backpropagation** is an efficient algorithm for computing the gradient of the loss function in a neural network.
- **Stochastic** gradient descent with minibatches allows efficient training even for very large training sets.
- **Adam** is an improved variant of SGD and is the algorithm of choice for training most neural networks. It scales the learning rate adaptively for every parameter.